

MASTER

Investigating process innovation adoption in the restaurant and construction industry a Theory of Planned Behavior approach

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Eindhoven University of Technology

Master Thesis

Investigating process innovation adoption in the restaurant and construction industry: a Theory of Planned Behavior approach

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Foreword

First I would like to thank my supervisors, Floor and Francesco, for guiding me through the process of writing my thesis. Special thanks to Francesco for being so supportive and flexible in your help, which meant that I could swing by pretty much any time, this was really appreciated. Second, I would like to thank my parents for being so supportive and always having my back anywhere anytime from day one. Next, I would like to thank my hermano, Stephan, never forget that moment when we decided to give studying our all, and we started pushing Pomodoro's till infinity, and I have never failed an exam since. I want to thank Nora for her support and positive vibes. I would like to thank my little brother, Mees, for all the humor over the past years, especially in the moments when I started taking things too seriously. Last but not least, I want to thank my older brother Tom for being my mirror and showing me how to take up a challenge with full commitment and conviction and never underestimate myself.

Summary

The Netherlands is dealing with labor market shortages that are predicted to remain a problem for the coming years as a consequence of an aging population and an already high labor participation rate. In addition to these shortages, labor productivity, which is the measure of economic performance that compares the amount of goods and services produced (output) with the number of hours worked to produce those goods and services, is decreasing. A potential solution lies in the utilization of process innovations to increase the labor productivity of the current workforce, and thereby decrease the need for more workers. Process innovation is defined as improving processes through the development and/or implementation of new technologies and/or methods for the company, thereby increasing its productivity and/or achieving broader societal/social objectives. Examples of process innovations include the implementation of digitalization, automation, and robotization. Unfortunately, it is unclear why this solution remains underemphasized in the action perspectives of employers.

This research addresses this issue by investigating two distinct sectors: the restaurant industry and the construction industry. Both industries face labor market shortages, have potential process innovations to implement, and are known for their conservative natures, yet have completely different values that they sell to their customers. Furthermore, since there is a lack of understanding of why employers have thus far underemphasized the option to invest and implement process innovations, this research adopts a psychological framework to analyze the underlying motivations, barriers, and drivers that influence the adoption or rejection of these process innovations. The chosen psychological framework is the Theory of Planned Behavior (TPB) developed by Icek Ajzen in 1985. The target group interviewed to gain evidence are employers or decision-makers, from the companies in the chosen sectors, that have the role or task of deciding to adopt process innovations. The main research question of this study is:

What are the factors that influence the intention and decision to implement process innovation among decision-makers in the restaurant industry and construction industry?

This research question was answered in a series of steps. First, a literature review (Chapter 2) was performed to get an overview of the possible process innovations to adopt for each sector. This literature review then investigated the barriers to the adoption of these process innovations. Secondly, the barriers from the literature review were combined with the TPB to create the conceptual model of the barriers that influenced the individual's attitude, subjective norm, and perceived behavior control (PBC). The conceptual model served as the structure on which the interview protocol is based. Thirdly, 21 semi-structured interviews were performed and transcribed (Chapter 3). These transcriptions were analyzed using both a deductive and inductive approach, to find both barriers and drivers that influence the individual's attitude, subjective norm, and PBC to either increase or decrease the likelihood of the adoption of process innovations (Chapter 4). Fourthly, both the common and differing factors for the adoption of process innovations in the two industries are discussed, based on the results from the barriers and drivers (Chapter 5). This master thesis ends with a conclusion of the findings and a policy recommendation (Chapter 6).

The literature review obtained the following barriers for the restaurant industry; *lack of trust* in technology influencing the attitude, *resistance to change* from employees and customers influencing the subjective norm, and *infrastructural barriers, lack of human capital*, and *upfront financial costs* that influence the PBC. For the construction industry, the literature review identified *resistance to change* from employees influencing the subjective norm, and *lack of technology awareness, lack of skills*, and *cost of implementation* to influence the PBC.

Results

From the semi-structured interviews, the following barriers to adoption were identified in the restaurant industry: *disruption or no added value* to the restaurant shaping a negative attitude toward the adoption

of process innovations. Secondly, *resistance to change* from employees and customers influences the subjective norm. Finally, *Financial costs*, *infrastructural barriers*, *human capital resources*, and *technology maturity* negatively influenced the PBC of decision-makers. On the contrary, the following drivers for the adoption of process innovations in the restaurant industry were identified; the *Innovator Mindset* describes a positive attitude toward the adoption of process innovation because the innovation functions as an enabler of the decision-maker's goals. The *Benefits* of increased job satisfaction among employees, productivity increases, and commercial value that come with the adoption of process innovation positively influence the decisionmaker's attitude. *External shocks* in the form of COVID-19 and labor market shortages influence the PBC by putting pressure on the decision-maker to adopt process innovations. Finally, *Alternative acquisition structures* can solve the high upfront implementation cost barrier thereby positively influencing the PBC of the decisionmaker.

For the construction industry, the interviews identified the following barriers; the *Short-term perspective* which is illustrated by a project-based approach which creates a lack of predictability and standardization, shaping a negative attitude toward the adoption of process innovations. Secondly, the *Resistance to change* from employees and other partners in the construction chain, referring to subcontractors or clients, negatively influences the subjective norm of the decision-maker. This resistance to change can be increased by a lack of technology awareness. Finally, the *Cost of implementation* and *Lack of skills* influencing the PBC of the decision-maker was also confirmed in the interviews. In contrast, the following drivers for the adoption of process innovations in the construction industry were identified; A *Long-term perspective*, illustrated by a project-transcending approach and a focus on company continuity shapes a positive attitude toward the adoption of process innovations. Secondly, *Benefits* in the form of increased quality and material efficiency, as well as, time savings and cost reductions also shape a positive attitude toward the adoption of process innovations. Thirdly, the *Innovator Mindset* was also identified in the construction industry. Fourthly, *Demand-based* drivers in the form of legislation from the government, or customer demands can influence the subjective norm of the decision-maker into the adoption of process innovation. Finally, *External shocks* in the form of labor market shortages to influence the PBC of the decision-maker were also identified in the construction industry.

Conclusion

From these barriers and drivers, the main factors were identified influencing the adoption of process innovations, four of which applied to both industries and are therefore generalizable to other industries, and two specific to the construction industry. The main factors that apply to both industries are the following *Innovator Mindset*, *Necessity*, *Resources and Opportunities*, and *Applicability*. The two other factors that were only applicable to the construction industry are the *Time Horizon* and the *Human Capital* factor.

In both the restaurant and construction industries the adoption of process innovations is influenced by the following factors. First of all, some decision-makers have the *Innovator Mindset* attitude, which describes a certain positive attitude toward the adoption of process innovations because they are constantly looking to improve existing processes. The benefits that originate from the adoption of process innovations, such as increased productivity, quality, and time savings are in line with this constant search for improvements, which thereby increases the likelihood of the adoption of process innovations. Secondly, the *Necessity* to adopt can be increased by external pressures that drive decision-makers toward adopting process innovations, for example, because of COVID-19 or labor market shortages. In addition, *Necessity* can also be increased by factors that influence the subjective norms of decision-makers in the form of legislation or customer demands that require the adoption of certain process innovations. In contrast, *Necessity* can be decreased by resistance to change from the employees of the company, or if the process innovation adds no value to the existing business operations. Thirdly, *Resources and Opportunities* play a direct role in influencing the PBC of decision-makers, which means

that barriers and drivers that influence the PBC can directly influence the decision-maker's ability to perform the behavior. Barriers that influence the PBC are for example cost of implementation, infrastructural barriers, or technologies that are unable to meet the company's requirements. Finally, *Applicability* refers to the process innovation being suitable to the type of company and its business operations. In the restaurant sector, for example, fast food restaurants have customers who are less bothered if the restaurants use robotization and digitalization in the creation of their meals. In contrast, restaurants that aim for a more personalized approach are less likely to use robots in their restaurants. Similarly, in the construction industry companies that are in the business of management and maintenance of buildings are more likely to use sensor-based technologies to create a digital picture of the performance of the building in terms of energy and comfort and display this to their customers through dashboarding. In contrast, new construction has a lot of potential for the implementation of BIM and prefabrication, as these companies have the chance to design the structure from a digital file up until the actual product.

Additionally, for the construction industry, the *Time Horizon* and *Human Capital* also influence the adoption of process innovations. The *Time Horizon* refers to the type of time perspective that the decision-maker has concerning the adoption of process innovation. A long-term perspective is more likely to lead to the adoption of process innovation because it incorporates the implementation challenges and places them in the perspective of a higher goal, such as company continuance. Finally, *Human Capital* relates to the specialized skills that are required for the adoption and implementation of process innovation in the construction industry. These highly skilled workers are often scarce in a company, therefore it is often difficult to build the capacity necessary for the implementation.

Policy recommendation

For the restaurant sector, based on the findings of this research, there seems to be no evidence that justifies a government intervention related to process innovations. First of all, no legislation creates a necessity for restaurant decision-makers to adopt process innovation, since no driving factors were found that influence the subjective norm of a restaurant decision-maker. Furthermore, the adoption of process innovation in the restaurant industry is mainly dependent on the type of restaurant and the suitability of the process innovation (*Applicability*), the attitude of the decision-maker (*Innovator Mindset*), and whether they perceive the process innovations to bring benefits to their restaurant. These are individual characteristics and abilities of the restaurant, which justifies no government intervention.

In contrast, the construction industry faces a multitude of legislation pressures from the government that originate in societal goals such as climate ambitions. These ambitions create the necessity that forces companies to change their existing processes. However, the government should be careful that it creates a growing necessity to adopt the process innovations necessary to adhere to regulations without companies having the financial and human capital resources to do so. The technician shortage will likely only increase in the Netherlands because younger people opt less for a technology-related education. This means that a large-scale transition might increase the necessity, while the bedrock of financial resources and human capital resources important to the success of these transitions is lacking. Especially since this research highlighted how the adoption of process innovations requires more workers who have specialized skills.

With a decline in productivity across sectors, increasing labor shortages, and the underutilized implementation of labor-saving process innovations, this study therefore argues that there should be a government focus on making sure there is money invested in skills development necessary for Construction 4.0 (C4.0) type process innovations in the Netherlands. This is especially applicable to small firms that are willing to adopt C4.0 type process innovations, but lack the financial resources and have trouble finding the necessary human capital. It is argued that government intervention is justified for increasing the human capital pool necessary for C4.0-type process innovations because it is part of the foundation of the government's transition goals.

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1. Introduction

On November 5th, 2020, a motion by members of the Dutch parliament was sent regarding the recognition that labor productivity is declining in the Netherlands and population aging will lead to labor market shortages in the long term (Amhaouch and Bruins, 2020). Labor productivity is *a measure of economic performance that compares the amount of goods and services produced (output) with the number of hours worked to produce those goods and services* (CBS, 2024). The relevance of this recognition lies in that a “broad economy and a strong manufacturing industry” serve as the bedrock of a strong and healthy economy (Amhaouch and Bruins, 2020). As a potential remedy to these problems, process innovation was mentioned to increase labor productivity, employment, and competitiveness for all regions and thereby play an important role in the future economic prosperity of the Netherlands. However, Amhaouch and Bruins (2020) noted that “process innovations are lagging in many small and medium-sized enterprise (SME) sectors”. In reaction to this motion, Zegel and Cloosterman (2021) wrote a report in which they explored the current situation regarding process innovation in the technical sector focusing mainly on the manufacturing industry and SMEs in the Netherlands. This was done to investigate whether a market or governmental failure justified governmental intervention. Process innovation is defined by Zegel and Cloosterman (2021) as “*Improving processes through the development and/or implementation of new technologies and/or methods for the company, thereby increasing its productivity and/or achieving broader societal/social objectives*”.

Zegel and Cloosterman (2021) proposed policy options for a suitable role for the government, ranging from no intervention or changes to the current situation to a completely new strategy based on economic, social, and societal goals, such as sustainability and labor market inclusion. The latter strategy transcends the question of only process innovation. On the 12th of October 2021, the Ministry of Economic Affairs and Climate decided to go for the option that entailed no intervention and changes to the policymaking surrounding process innovations (Blok, 2021).

This research posits that Dutch industries, including both SMEs and large firms, are still underutilizing process innovations to boost labor productivity which is a pressing concern in light of increasing labor market shortages. Therefore, this study will investigate the adoption of process innovations among Dutch companies in two sectors that deal with labor market shortages: the restaurant industry and the construction industry. This will be done by exploring both the barriers that hinder and the drivers that facilitate the adoption of process innovations, thereby providing more information on the factors that influence the decision-making processes within these companies.

1.1 Labor Shortages, Labor Productivity, and Process Innovations

Labor Shortages

In September of 2023, van Kesteren et al. (2023b) stated that “Almost all sectors in the Netherlands are experiencing labor market shortages”. This problem manifests itself both quantitatively and qualitatively. Quantitative in that organizations have difficulties filling in the vacancies, and qualitative in the sense of having the appropriate skills for the job at hand (van Kesteren et al., 2023b). The shortages likely remain a problem for the coming years since population aging is predicted to increase and the current labor participation rate of the Netherlands is already high (van Kesteren et al., 2023b).

The consequences of these labor shortages are manifold. Firstly it will cause an increase in the work pressure for current employees (Heyma et al. 2022; van Kesteren et al., 2023a). This will mean that less work can be finished and that delivery times for consumers will increase. The costs for companies will increase as they might have to outsource more work, decreasing the revenue streams. In the long run,

this labor shortage means that there will be less time for education, mentoring, and (re)training of employees. Heyma et al. (2022) explain that this will have negative consequences for the quality and continuity of the sectors.

Secondly, specifically, technician shortages threaten the slow climate ambitions of the Dutch government, referring to the energy and sustainability transitions and digitalization (Heyma et al., 2022). With the ambitious climate goals set by the Dutch parliament in the 2021 coalition agreement, the Netherlands should be CO₂-neutral by 2050 (Adriaansens et al., 2023). Leaving aside all the required changes in technologies, procedures, and legislation the most pressing barrier might be that there are not enough people to execute the necessary labor (Vleugels et al., 2022; van der Leij, 2023). In addition to the climate ambitions, the Netherlands is also dealing with a housing shortage, which instead of decreasing is still increasing (Naaktgeboren and Klerks, 2024). Even though existing construction projects for new houses face financial difficulties due to high interest rates, the construction sector is estimated to have 51.000 vacancies in 2024 (Mollemans, 2024). This increase in labor demand originates from the restoration, maintenance, and renovation of existing buildings, which is less dependent on economic developments in comparison to new construction projects.

Thirdly, next to these societal consequences labor shortages will also cause economic impacts, as they affect technological development and innovations, which are an important source of economic prosperity (Heyma et al., 2022). The annual questionnaire survey that focuses on innovation and entrepreneurship among Dutch companies, called the Dutch Innovation Monitor, stated that shortage in the labor market was one of the key macroeconomic disruptions in the value chains of companies (de Jong et al., 2023). Complementarily to this research, Buck Consultants International (BCI) concluded that access to talent, or the lack thereof, was mentioned as a pull factor for companies to move their businesses abroad (BCI, 2023. P. 22-23). The companies that were interviewed in the study mainly saw problems in the accommodation of growth in the Netherlands, because of a lack of supply of talent and the possibility of housing these talents, especially in the high-tech industry (BCI, 2023, p. 1). BCI states that this does not mean that existing business activities in the Netherlands are moved abroad, but that new investments are more likely to be made abroad. This will have consequences for the Dutch economy in the long run.

Labor Productivity

Given that the Netherlands cannot count on a larger workforce to address these future challenges, an alternative solution involves enhancing the productivity of the existing workforce through improvements in labor productivity.

Labor productivity can be influenced by three production factors (van den Berg et al., 2022). Firstly by investing in capital, for example, a company invests in an automatic bricklaying machine which allows for faster bricklaying increasing the amount of work done per hour. Secondly, labor productivity can be increased by increasing the training and education of existing workers so that they become more experienced. Then there are also changes in output that cannot be assigned to either of these two, which refer to multifactor productivity (MFP) (van den Berg et al., 2022). MFP increases are often linked to innovation and technological progress, and often fundamentally change the existing processes. For example, the scenario where a construction company implements Building Information Modeling (BIM) software across its projects. The technology does not just make the existing process faster, it changes how different factors, such as labor, material, and machinery are coordinated and utilized, effectively changing the construction process.

Unfortunately, both labor productivity and MFP have been decreasing in the Netherlands in the past few years, and this has been the same for most developed countries (van den Berg et al., 2022). This is part of a larger story often called the “productivity paradox” (van den Berg et al., 2022), which concerns

the co-existence of exciting new radical technological innovations and declining productivity in OECD economies (Fragkandreas, 2021).

Process Innovation

That brings us back to recent times, where three reports investigating the labor market shortages have concluded that one of the solutions that remain underemphasized in the action perspectives of employers is the possibility of investing in and implementing labor-saving process innovations such as automation and robotization (Heyma et al., 2022; van Kesteren et al., 2023a; van Kesteren et al., 2023b).

There is a positive correlation between the implementation of automation and growth in labor productivity (van Kesteren et al., 2023a). It is therefore that, Van Kesteren et al. (2023b) believe that the Netherlands could increase its labor productivity substantially through better implementation of automation and robotization of labor processes. However, with increasing pressures from labor market shortages and societal goals, employers still underemphasize the option to invest and implement labor-saving process innovations to increase their productivity and decrease their need for additional workers. The question thus remains why is this the case?

1.2 Focus of this Research

This research addresses this question by investigating two distinct sectors: the restaurant industry and the construction industry. Both sectors face labor market shortages, have potential process innovations to implement, and are known for their conservative natures, yet have completely different values that they sell to their customers. Furthermore, since there is a lack of understanding of why employers have thus far underemphasized the option to invest and implement process innovations, this research adopts a psychological framework to analyze the underlying motivations, barriers, and enablers that influence the adoption or rejection of these process innovations. The chosen psychological framework is the Theory of Planned Behavior (TPB) developed by Icek Ajzen in 1985 and will be further explained in the literature section. The target group interviewed to gain evidence are employers or decision-makers, from the companies in the chosen sectors, that have the role or task of deciding to adopt process innovations.

This research uses the definition of process innovation as given by Zegel and Cloosterman (2021), and thus includes a wide variety of technologies that relate to the adoption of automation, digitalization, robotization, and artificial intelligence (AI), with the goal of increasing productivity and, or achieving broader societal goals. Examples of process innovation in the construction industry could be the implementation of BIM technology to digitalize the planning, design, construction, and management of buildings. In the restaurant industry, an example is the implementation of a digital ordering system, such as QR codes that can be scanned on phones for ordering food and drinks.

1.2.1 Research Question and Sub-Questions

The main *research question* of this study is:

RQ: What are the factors that influence the intention and decision to implement process innovation among decision-makers in the restaurant industry and construction industry?

The Sub-questions are:

SQ1: What are the barriers that shape the attitude, subjective norms, and perceived behavioral control away from the adoption of process innovations?

SQ2: What are the driving mechanisms that shape the attitude, subjective norms, and perceived behavioral control toward the adoption of process innovations?

1.2.2 Sectors

The first sector that will be included is the restaurant sector. In the Netherlands, some sectors perform better than others in terms of labor productivity. However, Van Kesteren et al. (2023a, p.31) named the hospitality industry as one of the top three sectors that lacked productivity growth between 2009 and 2019, with a productivity decline of -13%. Santini et al. (2021) highlight similar findings in the US, where the restaurant business had one of the lowest labor productivity among different industries, with a stagnation in labor productivity growth in the years before 2015.

The second sector that will be included in the study is the construction industry. This industry includes construction, renovation, and installation businesses. The construction industry has been chosen because next to dealing with the challenges mentioned earlier, it has also been tasked with adhering to societal goals such as building more homes to decrease the housing shortage and doing so in a sustainable fashion. Now the productivity of the construction industry has been lagging for years according to an article by van Sante (2022), in which he compares the construction industry to that of manufacturing. Van Sante (2022) explains that the labor productivity of construction is significantly falling behind that of manufacturing. Over the past 25 years, labor productivity in manufacturing has nearly doubled, allowing a manufacturing worker to produce twice as much in the same amount of time as a construction worker (van Sante, 2022). To deal with labor shortages and ensure that prices do not rise too much “further digitalization is essential” (van Sante, 2022). In a similar fashion to construction, the installation sector will also play an important role in successfully achieving societal goals, look for example at the European Union’s goal of installing 60 million extra heat pumps by 2030 (Nuttal et al., 2023). These heat pumps will play an important role in decarbonizing the heating of European homes, an important aspect of the overall climate ambitions. However, one of the significant barriers to this goal, is that “companies in Europe, including in the heat pump sector are struggling to recruit skilled electricians, technicians, engineers and construction workers” (Nuttal et al., 2023, p. 8).

It is important to recognize that the construction industry consists of a diverse range of company types, such as those building new structures, but also firms that install a variety of installation systems such as electrical systems, plumbing, heating, ventilation, and air conditioning (HVAC). These companies differ in their value propositions and business operations and thus also in their decision-making toward adopting process innovations. Consequently, the analysis of the construction industry will include a wider variety of companies and decision-makers such as owners, directors, (innovation) managers, and project developers. In contrast, the restaurant industry is characterized by more uniform value propositions and business operations and will include restaurant owners, managers, and head chefs.

1.3 Outline

This thesis is continued as follows; Chapter 2 will provide a conceptual model of the barriers to adoption of process innovations for each sector based on a literature review. It does this by first, giving an overview of the possible process innovations to adopt for both the restaurant and construction industry. Succeeding this overview the literature review continues by highlighting the barriers to the adoption of these process innovations per sector. Chapter 2 then continues to explain the theoretical framework, the TPB, used for creating the conceptual model of this research, which serves as the structure on which the interview protocol is based. This section will furthermore elaborate on the suitability of using the TPB for this research, thereby also highlighting some applications and criticisms of the TPB. Chapter 3 will discuss the methodology of this research, explaining the qualitative research design and discussing the sampling strategy, data collection, and data analysis methods used for analyzing the interview transcriptions. The results of this qualitative analysis will be highlighted in Chapter 4, from

which the first and second sub-questions are answered. Chapter 5 will delve into a discussion of these results, highlighting both the common factors influencing the adoption of process innovations in both industries and the factors that differ per industry. Embedded in this discussion are the similarities and differences between the conceptual model obtained at the end of Chapter 2 and the results from the interviews performed in this research. Chapter 5 continues by discussing the generalizability of the factors to other industries, the limitations of this study, and potential directions for future research. The final chapter will provide a brief conclusion of the most important findings of this study and end with a policy recommendation.

2. Literature Review

Before this literature review goes into detail about the two sectors, it will first briefly highlight the barriers to process innovation that were mentioned by the study of Zegel and Cloosterman (2021) which investigated process innovation in the manufacturing industry and SME sectors in the Netherlands as discussed in the introduction. Zegel and Cloosterman (2021) found that a lack of investments in process innovations is often not rooted in a lack of available innovations, but more so in the implementation thereof. Technologies and methodologies for implementation are generally available, however implementations face challenges with internal business processes. Zegel and Cloosterman (2021) identified the following barriers; companies experience a lack of urgency to invest in process innovations and focus on the short term. Secondly, companies lack the knowledge necessary to obtain a business case. Thirdly, they struggle to create internal capacity, meaning freeing up the right people for the process innovation. Fourth, they face issues concerning the acceptance of change, and finally, some companies lack the required financial resources. This section will continue by investigating the restaurant and construction industry, and their specific barriers to the adoption of process innovations.

2.1 Literature Review Restaurant Business

Innovation in the restaurant industry has not stayed constant in the last decades. To illustrate, Oronsky et al. wrote in 2007 that “the average full-service restaurant firm would not be oriented towards incorporating new technology into operations and functional aspects of the business until such time that the technology becomes a necessity for survival”. In contrast, Kim et al. (2023) concluded that “automated food preparation, table order, and delivery management system are necessary for process optimization in the kitchen and restaurants”. These two statements highlight the most extreme ends of an innovation strategy spectrum for the restaurant industry. In the middle of the two extremes are conclusions drawn by Lee et al. (2019), which showed that restaurant entrepreneurs mainly look at competitors for their innovative ideas. That is, the most common form of innovation is that of the imitation of successful others, listening to customers, and scanning media sources thereby reducing the risks and uncertainties that are involved in the development of innovations (Lee et al. 2016; Lee et al., 2019).

It would be interesting to understand how these different perceptions of process innovation in restaurants change when pressing labor shortages restrict the ability to hire personnel. The innovation climate in the restaurant sector might change, and there are already indications of its potential. Guan et al. (2022) highlighted that robotic hotel and restaurant implementations are becoming more common, even though such advancements look futuristic. Many repetitive or time-consuming operations in restaurants can be automated by robots, increasing productivity and facilitating efficient service (Zhu, 2022). By improving productivity, Grobbelaar et al. (2021) argued that the implementation of robots could solve the shortage of labor supply. In a recent study exploring customers’ acceptance of robots, artificial intelligence, and service automation (RAISA), Santiago et al. (2024) concluded that restaurant owners can start implementing new technologies as customers are ready to accept RAISA implementations in dining settings.

To innovate a process, however, one must understand where innovation possibilities are present. The next section highlights examples of prior process innovation implementations in the restaurant industry.

2.1.1 Examples of process innovations in the restaurant business

Kim et al. (2023) identified three necessities for process optimizations in the kitchen and restaurants, namely automated food preparation, delivery management system, and automated table ordering.

Automated food preparation can be understood as all potential automation implementations available to support or take over the process of preparing food. Examples include Spyce Robotic Kitchen in New York which utilizes robots to prepare affordable healthy meals, as customers are not always looking for an experience while dining (Grobbelaar et al. 2021). Also, Big Eyes Sushi is a restaurant in Brooklyn that employs robots for sushi preparation (Santini et al. 2021). “The goal of the robots is to increase speed, efficiency, and consistency, which can make up to 60 sushi rolls per minute” (News 12 Staff, 2019).

With a delivery management system, one can think of an action that involves the movement of food toward the customer or table, as well as dirty plates back to the kitchen. Robotic waiters can play a role in this process and can vary in the complexity of tasks that they can handle (Berezina et al. (2021). Some robotic waiters are limited to obstacle detection and can only deliver food through the restaurant (Berezina et al., 2021). An example of such a support robot is an Automated Guided Vehicle (AGV), which helps in serving food and bringing back dishes to the kitchen. Shimmura et al. (2020) performed a study using the implementation of an AGV system and showed that it resulted in reduced work hours and the enhancement of labor productivity. Other robots may have more complex abilities such as anthropomorphic features that allow for limited communication with customers and therefore engage in more complex interactions (Berezina et al. 2021).

Concerning automated table ordering, one can think of the scanning of QR codes at the table, which automatically sends the order to the kitchen (Santiago et al., 2024). More complex innovations in this area could involve research performed in RAISA. Santiago et al. (2024) discuss the work of Nozawa et al. (2022) in which they explain how “AI-based dining management systems may enhance the eating experience by identifying table occupancy, producing real-time table status diagrams, increasing restaurant availability rates, and enhancing the effectiveness and quality of the services provided”.

From the applications described above it becomes evident that there are different options or routes to pursue the adoption and implementation of process innovation. So far this is consistent with the findings of Zegel and Cloosterman (2021. p. 2), who stated that on average the availability of innovations is not the problem it is the implementation thereof. The question thus remains which factors play a role in the decision to implement or obstruct the implementation of process innovation in the restaurant industry?

2.1.2 Barriers and challenges to the adoption of process innovations in the restaurant business

Although process innovations such as robotization and digitalization are gaining popularity, as already mentioned in the introduction, the restaurant business as a whole stagnates in terms of labor productivity, indicating that a majority of restaurant owners choose to not invest in process innovations. This section tried to identify the barriers found in the literature, and five main barriers became evident.

Financial costs

The first barrier, as explained by Ivanov and Webster (2017), can be summarized under the term financial costs. Ivanov and Webster found that both the purchasing, installation, and maintenance costs form a barrier to the implementation of robots or automation options in restaurants. It is interesting to note that Oronsky et al. (2007) argued that “the absence of formal capital budgeting techniques might explain both why independent as well as full-service restaurants demonstrate a lack of technology leadership.

Infrastructural barriers

The second barrier relates to infrastructural barriers. The implementation of a robot or automation might require significant changes to be made to the restaurant, for example, robots might require ramps,

special tracks, or platforms (Ivanov and Webster, 2017). Apart from physical barriers, there could also be digital barriers. For digitalization implementation such as smart devices, or smart robots, that use the Internet-of-Things (IoT), Pappas et al. (2021) stressed the necessity for a sufficient network infrastructure to support IoT.

Resistance to change

The third barrier originates from the work of Lee et al. (2019), which argued that the employees' unwillingness to change as well as customer non-acceptance of innovations, form barriers to the generation and implementation of new ideas. Consumer resistance to RAISA seems to be dependent on the type of restaurant. For example, for luxury restaurants customers pay extensive amounts of money and value human service (Nozawa et al., 2022). In contrast, when people are looking for a quick meal (Grobbelaar et al., 2021), or are just more accepting of technology in general (Santiago et al. 2024), robotic implementations can be very beneficial.

Trust

The fourth barrier is the trust between technology and humans. Trust seems to be underdeveloped in Human Robotic Interaction, this lack of trust in industrial robots or cobots results in the refusal to engage in robot cooperation (Grobbelaar et al, 2021). The customer's lack of trust in robots mainly originates from their questioning of the robot's ability to follow strict safety rules (Grobbelaar et al., 2021). An example is that kitchen automation is associated with inflexibility, which means that dishes cannot be customized. This may create perceived risks when handling specific food allergy requests.

Lack of human capital

The final barrier relates to a lack of resources in the human capital domain. Weerasinghe and Nirere (2022) found in their qualitative research that restaurants often lack the technical knowledge related to the technologies implemented. Results for example showed that restaurant owners sometimes needed to wait hours for technical assistance in times of technological errors (Weerasinghe and Nirere, 2022). This stresses the need to also obtain the required skills among employees or employers to operate the new process innovation.

2.2 Literature Review Construction Industry

This section discusses the innovation climate in the construction industry. The construction industry is infamous for its slow adoption pace of innovations (Newman et al., 2021). But with recent developments in societal objectives, such as that of future-proof living environments, the construction industry requires new employees with the right skills and a cultural change (van der Maarel 2023). The changing horizon through developments in automation, sustainable energy, urbanization, and ICT in the living environment, forces the construction industry to change its existing approaches and processes (Volberda and Heij, 2019). This is especially true since the Netherlands is dealing with a technician shortage, which is predicted to remain a problem in the coming years, because of population aging and an already high labor participation rate (van Kesteren et al., 2023b, p.49).

The implementation and discovery of process innovations in the construction industry can be considered as a subset of the broader developments that fall under the umbrella term of Industry 4.0 (I4.0). McKinsey and Company (2022) describe I4.0 as “the next phase on the digitization of the manufacturing sector, driven by disruptive trends including the rise of data and connectivity, analytics, human-machine interaction, and improvements in robotics”. Siriwardhana and Moehler (2023), describe how the construction industry has started to implement I4.0's ideas in the form of for example digitization and automation, thereby creating the development of Construction 4.0 (C4.0). C4.0 can bring many benefits to the industry, such as reducing labor and material costs, while increasing

efficiency, performance, and collaboration (Siriwardhana and Moehler, 2023). The next section will discuss examples of these types of process innovations.

2.2.1 Examples of process innovations in the construction industry

This section discusses a variety of process innovations in the construction industry such as construction and installation branch. The literature investigated labor-saving process innovations and identified three main solution themes, namely prefabrication solutions, IoT solutions, and digital solutions. These three themes share an overlap with the advanced technologies that Newman et al., (2021) highlighted to play a role in C4.0: artificial intelligence, big data, Internet of Things (IoT), sensor-based technologies, 3D printing, cloud computing, advanced software such as Building Information Modelling (BIM), and cyber security.

Prefabrication

In the construction and installation sectors, prefabrication implementation seems to offer some relief from labor pressures. Prefabrication can be understood as the process of moving the assembly of products from the site of implementation to a place where the materials can be prefabricated more efficiently and in a controlled environment.

Prefabrication can play a role in gaining efficiency in the renovation of homes. Lihtmaa and Kalamees (2024) for example write that prefabrication can create time savings during renovations through activities such as insulation works, installation of windows, façade finishings, and installation of ventilation duct. Home renovations to make homes more energy efficient through for example better insulation plays an important role in achieving the goal of becoming carbon neutral by 2050 (Lihtmaa and Kalamees, 2024). In the Netherlands, the Ministry of the Interior and Kingdom Relations (Rijksoverheid, n.d.) has recognized that the renovation process is often deemed more sustainable, cost-effective, and environmentally friendly than the alternative option of demolition followed by new construction. Lihtmaa and Kalamees (2024) therefore call for an urgent need to improve resource utilization and overall construction efficiency by switching to a method that uses the prefabrication of construction elements.

Prefabrication could also be a time-saving innovation for the installation sector. For example in the installment of heat pumps, Rutten (2022) describes how for an innovative heat pump design, prefabrication was used to create a single unit to install. The unit includes a roof duct, meaning that the installer no longer requires a certification for gas appliances and the installment can be done in one day thus saving labor hours.

Internet-of-Things (IoT)

IoT is best understood as a complex system that consists of a diverse set of physical entities that are connected through the internet, allowing those entities to exchange and collect data and information (Trappey et al., 2017). These physical entities can be mechanical and digital machines, or other physical objects, that can perform real-world tasks without human activity. Sensor technologies can amplify the possibilities of IoT networks, as they can allow for the gathering of a wide range of information, such as physical quantities, qualities, or events, by for example measuring changes in environmental conditions such as heat, pressure, humidity, and movement (Trappey et al., 2017).

In the construction industry IoT can be used for example through installing connected sensors in buildings that gather real-time information regarding the structural health of the building and its

components (Ghosh et al., 2020). Another more advanced example involves the application of IoT integration with building information modeling (BIM) to create a digital twin of a building (Ghosh et al., 2020). This setup can be used to improve construction and operational processes. Ghosh et al. argue that IoT can provide the industry with new possibilities that could potentially increase its efficiency and thereby solve time and resource constraints.

Pomp et al. (2022; 2023) organized a hackathon to inspire and investigate different possibilities for IoT solutions in the construction industry. IoT makes use of different aspects of C4.0. In the hackathon employers from different craft backgrounds were invited. Since there is a lack of awareness of IoT in the construction industry, a barrier that will be discussed more extensively in the next section, Pomp et al. (2022; 2023) aim to develop an IoT platform for German craft businesses. This platform called IoT4H should enable businesses to first identify suitable use cases and resulting business models for their trade, and secondly, create an overview of the necessary components for implementation such as sensors.

To name some examples that the employers in the hackathon of Pomp et al. (2023) came up with. The first one could be the implementation of IoT to track tools or track the moisture penetration of a roof. IoT and sensors create a situation where increased availability of data allows craftspeople to gain a better understanding of situations and potentially provide predictive maintenance-type solutions. For example, the equipment of water gutters with flow sensors, allows for accurate tracking of the flow rate in the gutter after rainfall, thereby creating the ability to provide maintenance before a real problem occurs. These predictive maintenance-type solutions could also improve planning operations meaning that existing human capital could be used more efficiently.

Digital solutions

Volberda and Heij (2019) mention the concept of “mechanic 4.0”, describing various process innovations within the domain of mechanical work. For instance, they illustrate the operational procedures adopted by mechanics at ENECO Installatiebedrijven B.V., where daily assignments are digitally accessible through a tablet. The list of tasks remains dynamic throughout the day, allowing flexibility for unforeseen emergency repairs. Upon completing a task, mechanics are required to document the utilized parts, thereby enabling the quick identification of necessary reorders and updates to the inventory. Meanwhile, at Strukton Worksphere B.V., mechanics receive assignments and relevant details via a smartphone application.

Digital solutions can help in lowering the amount of site visits, by making the process more efficient. For example, when a tradesperson arrives at a task, different surprises can be encountered turning routine work into a complex assignment. Almond (2022) explains that video technology allows tradespeople to call in expertise in times of need. In another example, Almond (2022) describes how roofing professionals employ drones to quickly assess the work environment, enhancing their understanding of the situation. These types of simple technology implementations can save a lot of time and money.

The next section will highlight the main barriers that were identified for the adoption of the discussed types of process innovation in the construction industry.

2.2.2 Barriers and challenges to the adoption of process innovations in the construction industry

This section identifies four over-coupling barriers that obstruct the adoption of process innovations in the construction industry.

Cost of implementation

The first barrier is described by Newman et al. (2021) as the cost of implementation. This barrier has different aspects, and the first one is financial costs. Newman et al. describe that the UK construction industry has an average profit margin of around 1.5-2%, therefore investing in innovation can be perceived as too risky against the potential benefits. This perspective was supported by Lawani et al. (2022), who explained, using a case study of the Scottish bricklayer industry, that employers focus on maintaining financial profits, which investing in innovation decreases in the short term, therefore ignoring its implementation. Vazques and Garivani (2024) drew the same conclusion for timber construction projects, stating that initial capital requirements are high for digitally controlled machinery and robotics. Vazques and Garivani (2024) added that technologies that support productivity and ecological goals can only be adopted by large businesses with enough capital budget. This results in the important consequence that SMEs are either unable or unwilling to invest and participate in the digitizing economy. A second aspect of the cost of implementation barrier is that training employees can be expensive and time-consuming, thereby also decreasing short-term profit (Newman et al., 2021).

Resistance to change

Newman et al. (2021) describe that employees might be reluctant to adopt new technologies if they fear that it could eventually replace them or make their jobs redundant. It also does not help that the construction industry is conservative by nature, meaning that it is not known for its ability to embrace change (Demirkesen and Tezel, 2022).

Lack of awareness

The third barrier is best described as a lack of knowledge or awareness about innovations. Pomp et al. (2022; 2023) mentioned that the local skilled crafts, encompassing small and medium-sized enterprises, do not have the knowledge to evaluate the potential of and effectively develop and manage IoT solutions. Iturralde et al. (2022) made similar conclusions for the construction industry as they mentioned that the industry has a lack of technology awareness constraining the adoption of robotics and automation. To decide to invest in or implement an innovation, an employer needs to understand what the potential benefits are, Demirkesen and Tezel (2022) explain that benefits and gains are often not clear for C4.0-type technologies.

Lack of skills

The final barrier relates to the problem of lack of skills. Where the third barrier was more related to firms missing the awareness to see the potential of certain process innovations, lack of skills refers more to the inability to implement the innovation. Pomp et al. (2023) for example stated that small craft businesses have difficulty in “acquiring the knowledge and skills to develop and operate adequate IoT solutions”. After examining the theoretical constructs underpinning C4.0, Siriwarhana, and Moehler (2023) concluded that, despite the substantial productivity gains promised by C4.0, its widespread adoption often disregards the critical importance of specialized skills. This neglect will result in a disparity between expected outcomes and the actual results. Siriwarhana and Moehler stress that recent research on C4.0 places too much focus on technology while disregarding the importance of ability and skills development. These findings were also supported by the work of van der Heijden (2023), who investigated the implications of construction 4.0 on society and named a “shortage of skilled personnel trained in utilizing these technologies” as one of the key barriers. Newman et al. (2021) also recognized this and added another perspective on this barrier in that companies might hesitate to integrate new technologies into their daily activities. These companies contend that by using these technologies, production may suffer because of the need for highly qualified workers and a shift in the organizational culture.

The third and fourth barriers seem closely related yet have different implications for adoption. Lack of awareness for innovation does not directly say something about the willingness to implement innovation, but rather that one is not aware of possibilities. In contrast, a lack of skills and competencies restricts capabilities to implement process innovations. A solution that would benefit both barriers is an increased focus on training and development. Interestingly is that a literature review performed by Maqbool et al. (2024) concluded that inadequate training and development of employees played a major role in the skill shortage in the UK. This has resulted in decreased efficiency and lower-quality construction work.

2.3 Framework for analysis: Theory of Planned Behavior

This section will explain the theoretical framework that will be used to both structure and analyze the interviews that will be conducted for this research. The section will start with some historical background of the theory, after which the theory will be explained in-depth. This section will end with an explanation of the reasons for using this framework.

2.3.1 Historical Background

The Theory of Planned Behavior (TPB) is a psychological theory developed by Icek Ajzen in 1985. TPB aims to predict and understand human behavior. Fundamental to the theory is the assumption that intentions represent the motivational elements underlying a behavior, representing the level of commitment and effort people are willing to put into performing the behavior (Ajzen, 1991). The TPB argues that the intention directly precedes the behavior, and can be understood as a readiness indicator for performing that behavior (Ajzen, 2019a). The general rule holds that the stronger the intention to engage in a behavior, the higher the likelihood of the actual performance of the behavior should be.

TPB is an extension of the theory of reasoned action through the addition of perceived behavioral control. This addition addressed a significant limitation of the theory of reasoned action, which was that it did not control for situations where people had no or incomplete volitional control to perform a certain behavior (Ajzen, 1991). Non-motivational related factors can influence the adoption of a behavior. These non-motivational factors, which are called behavioral control factors, are related to the availability of necessary opportunities and resources such as time, money, and skills to perform the behavior (Ajzen, 1991, p.182). So the choice to perform a certain behavior depends on both motivational factors (intention) and behavioral control (ability and resources) as can be seen in Figure 1.

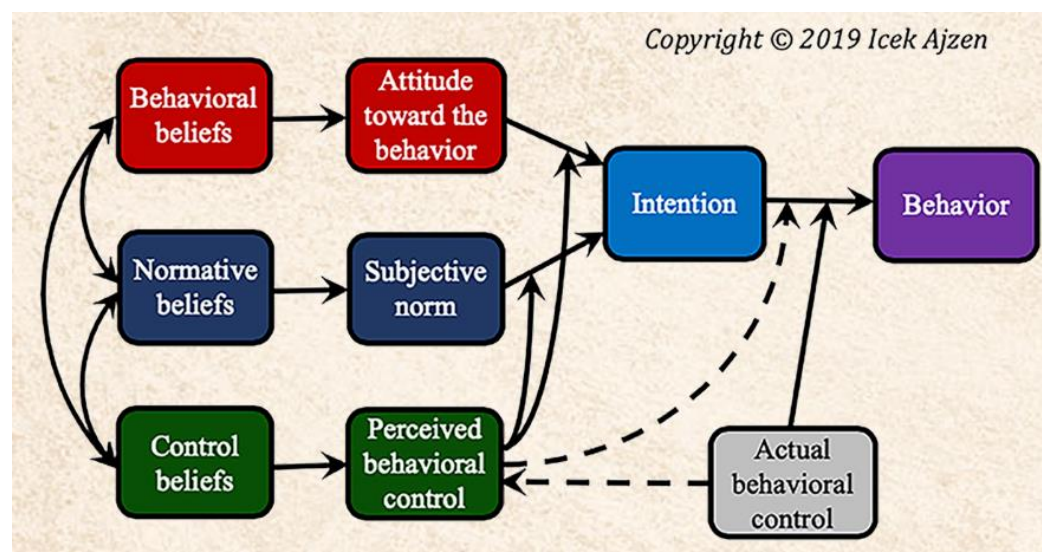


Figure 1 Overview of the Theory of Planned Behavior (Ajzen, 2019a)

2.3.2 Theory Explanation

The factors that influence someone's intention to perform a behavior are divided into three independent categories. The intention is directly based on the attitude toward the behavior and the subjective norm, both of which are moderated by the perceived behavioral control (Ajzen, 2019a). First, there is the attitude, which is defined as the extent to which an individual's assessment of the behavior in question is positive or negative (Ajzen, 1991, p. 188). The second factor is called subjective norm and resembles the social influence someone endures for performing the behavior in question (Ajzen, 1991). It refers specifically to the perceived social pressure to perform or not to perform the behavior. The third factor is called the degree of perceived behavioral control (PBC), which refers to the perceived ease or difficulty of performing the behavior and it is assumed to be influenced by prior experiences as well as expected barriers and difficulties (Ajzen, 1991, p.188). PBC closely resembles Bandura's (1982, p. 122) concept of perceived self-efficacy, which "is concerned with judgments of how well one can execute courses of action required to deal with prospective situations". PBC is important since the work of Bandura has shown that people's behavior is significantly shaped by their confidence in their ability to execute that behavior (Ajzen, 1991, p. 184).

Note that PBC can moderate the influences of attitude and subjective norm on intention, but in the case it accurately resembles actual control it can also moderate the effect of intention on the actual performance of the behavior. This means that an intention that is willing to perform the behavior can only actually do so if the PBC allows it to (Ajzen, 2019a). To illustrate, a decision-maker in a construction contractor firm might have a positive attitude toward the adoption of a bricklaying robot, and all employees in the company are in favor (positive subjective norm) meaning that the intention to adopt is strong, yet the company does not have the financial resources to purchase the robot (low PBC), meaning that the actual performance of the adoption will not happen.

Furthermore, TPB has the goal of explaining human behavior, not only predicting it (Ajzen, 1991). This means that the TPB also takes into consideration the antecedents of attitudes, subjective norms, and perceived behavioral control. These antecedents to the independent categories are determined on the basis that "behavior is a function of salient information, or beliefs relevant to the behavior" (Ajzen, 1991, p. 189). TPB argues that these salient beliefs are the most prevalent factors in a person's intentions and actions because TPB is built on the assumption that at any given moment humans have access to only a limited amount of information (Ajzen, 1991). TPB distinguishes three different forms of salient beliefs. The behavioral beliefs are assumed to shape attitudes toward a behavior, normative beliefs form the basis of subjective norms, and control beliefs lay the groundwork for perceptions of behavioral control (Ajzen, 1991). The following section will discuss the beliefs that form the foundation of the three categories that shape the intention while also giving examples related to the adoption and implementation of process innovations. An overview of the TPB framework can be seen in Figure 1.

The formation of attitudes: behavioral beliefs

Behavioral beliefs shape our attitudes, but what is the mechanism behind this? Ajzen (1991) uses an information-processing approach to the formation of attitudes, arguing that attitudes are developed from the beliefs people have about the object of the attitude. The beliefs about an object are formed based on the association we have with the attributes of the object such as its characteristics, events, or other objects. In the case the object is the performance of a behavior, the behavioral belief links the behavior to an expected result of performing the behavior, referring to the expected outcome or experience. A behavioral belief is the subjective probability that the behavior will produce a given outcome or experience (Ajzen, 2019a). Since the subjective outcome of the behavior is already labeled positive or negative, the person automatically obtains an attitude toward that behavior in line with the evaluation of that subjective outcome. In this way, behaviors that lead to a positive subjective outcome are met

with a positive attitude, and behaviors that lead to a negative subjective outcome with a negative attitude (Ajzen, 1991).

To apply this to an example, imagine the decision-maker of a roofer company is considering the adoption of drones for aerial site inspections. The decision-maker forms beliefs about this innovation by associating it with various outcomes. If the decision-maker links the adoption of drones to that it will result in reduced labor costs and time savings (outcomes), that is the outcomes of the behavior are both valued positively, this will likely result in a positive attitude toward the adoption of drones. However, in the case the decision-maker links the usage of the drone to complexities surrounding its implementation such as decreased profit margins (outcome) because of high initial investment costs or extensive training requirements for employees which take a lot of time, the attitude toward adoption will be labeled negative.

The formation of subjective norms: normative beliefs

Subjective norms are shaped by the interaction between normative beliefs and the motivation the individual has to comply with the people who hold them. Normative beliefs relate to the perceived viewpoints of influential others, known as “referent others” (Ajzen, 1991). Normative beliefs encompass the individual’s understanding of how these referent others view and evaluate a specific behavior. Motivation to comply is the extent to which the person wants to adhere to the preferences of these referent others (Ajzen, 1991).

In the application to adopting process innovation, take for example a restaurant owner who is considering the implementation of a new kitchen robot. The restaurant owner might feel that the employees do not approve of the implementation of robots, because they fear it will take over their jobs. This would be a normative belief. The relevant motivation to comply is the importance the owner attaches to the opinions of the employees. Again the two interact, so even if the owner feels that their employees would not approve of the robot implementation, this would not impact their intention to implement the robots if the owner does not care about the opinion of their employees.

The formation of perceived behavioral control: control beliefs

Finally, according to TPB the beliefs that ultimately determine whether the behavior can be performed are related to perceived behavior control, which refers to a person's perceptions of the ease or difficulty of performing the behavior (Ajzen, 1991). The PBC is determined by control beliefs which refer to the perceived presence or absence of factors, such as resources and opportunities, that may enhance or obstruct the performance of a behavior thus influencing its perceived difficulty (Ajzen, 1991).

These control factors can be situational, such as having access to financial resources, as well as personal, like having the knowledge and skills to operate a new robot system in the restaurant. Suppose, in the example above, that the implementation of serving robots requires access to a financial loan. Yet the restaurant owner has never requested a financial loan before and has limited knowledge of how and where to get one. The control belief of the restaurant owner about the financial loan (one of the resources required to perform the implementation of robots) would be low, thereby increasing the perceived difficulty of the adoption and thus decreasing the perceived behavioral control. This will likely result in the restaurant owner not adopting the process innovation.

As a general rule the more positive the attitude and the subjective norm are concerning the behavior, and the greater the perceived behavioral control is, then the individual’s intention to perform the behavior in question should increase.

2.3.3 Reasons for using TPB and Criticisms

Suitability of TPB to study adoption behavior with qualitative research

To analyze what influences the adoption of process innovation in the specified sectors, the Theory of Planned behavior (TPB) was adopted for different reasons. First, TPB was designed to predict behaviors across many settings (Mathiesson, 1991), and its diverse structure provides a solid framework to analyze the different aspects that could play a potential role in adopting process innovation. To illustrate the wide applicability of TPB, it has been used in more than 4,200 papers referenced in the Web of Science bibliographic database, making it one of the most applied theories in the social and behavioral sciences across a wide variety of research domains (Bosnjak et al., 2020). These domains include among others, business, environmental sciences, management, hospitality leisure sport tourism, and public environmental occupational health (Bosnjak et al., 2020).

TPB has also frequently been used to study the adoption of technologies across different industry settings in both qualitative and quantitative fashion. To highlight a few quantitative examples, Morris and Venkatesh (2000) investigated age differences in individual adoption and sustained usage of technology in the workplace using the TPB. Scannell et al. (2011) used the TPB framework to identify the key factors that influence the adoption of different shop floor manufacturing technologies. The TPB was shown to be an effective predictor of technology adoption in this context (Scannell et al., 2011). In a more recent study, Ivanov et al. (2024) investigated the drivers of generative AI adoption in higher education using the TPB and found that the TPB categories significantly influenced lecturers' and students' intentions to use generative AI tools. The intention in turn also significantly influenced the adoption of these tools (Ivanov et al., 2024).

TPB can also be used in qualitative research, as Ajzen himself states "TPB can be used as a heuristic framework to guide questions to be raised in qualitative research" (Ajzen, 2019b). To highlight examples of the qualitative application of TPB for adoption research, Chidembo et al. (2024) applied the TPB framework using interviews to investigate the underlying beliefs that influence the adoption of a solar home system in the Vhembe district municipality in South Africa. Renzi and Klobas (2008) used semi-structured interviews based on the TPB to explore the factors that influence the adoption of teaching models among university teachers. Finally, Monteleone et al. (2024) performed expert interviews based on the TPB to investigate the factors that can influence the adoption of innovations related to precision agriculture.

TPB is not the only behavioral theory that can be used to study the adoption of process innovations. Other models such as the Technology Acceptance Model (TAM) by Davis (1985) could be used for similar questions. However, the main reason TPB was chosen over TAM is twofold. First TPB does a better job of capturing the social aspects that play a role in adopting a technology (Mathieson, 1991). The other reason is that TPB does a better job of assessing the influence of behavioral control on the adoption of an innovation. TAM only uses the variable "ease-of-use" for behavioral control, while TPB divides perceived behavioral control into both internal and external control factors. Internal control factors relate to the skills and capabilities of the person to perform the behavior, while external control refers more to resource factors such as time and money (Mathieson, 1991). TPB thereby allows for a more in-depth investigation into which factors play a role in the intention to adopt a certain technology.

Criticisms of TPB

Although the wide application of the TPB across different research fields, the TPB has been criticized for several limitations. Firstly, Sheeran et al. (2013) criticized the TPB for ignoring unconscious influences on behavior and focusing solely on rational reasoning. Conner et al. (2013) similarly argued that the TPB overlooks how emotions influence behavior, which influences go further than only those anticipated from expected outcomes. Thirdly, the static nature of the TPB fails to account for how observed behaviors can influence cognitions and therefore subsequent behaviors (McEachan et al.,

2011; Sutton, 1994). In addition, Sniehotta et al. (2014) argue that the shortcomings of the TPB are illustrated by the researchers' frequent use of extended versions of the TPB. For example, Cobelli et al. (2021) investigated the factors that affect the choice to adopt or not adopt telemedicine technologies, using an extended version of the TPB. The extension was used because the researchers wanted to gain a better understanding of the underlying mechanisms that shape attitudes toward the adoption of telemedicine (Cobelli et al., 2021). This was done by including the underlying performance expectancy, anxiety, and self-efficacy that shape the attitude. A similar type of extension to the TPB framework was done by Pavlou and Fygenson (2006) in their study to understand and predict electronic commerce adoption. Pavlou and Fygenson (2006) split e-commerce adoption into getting information (from the website of the vendor) and the intention to purchase and therefore extended the TPB to predict both behaviors. The extension meant that the antecedents of PBC, self-efficacy, and controllability were included in the model. This made sure that both the individual's capabilities were measured as well as controlling these for the availability of resources and opportunities to execute the behavior (Pavlou and Fygenson, 2006).

The reason to use TPB despite the aforementioned criticism and limitations is first of all that the adoption of a process innovation often comes with considerable upfront implementation costs and time necessary to incorporate it into existing business processes, resulting in the decision to adopt process innovations is more likely to be made deliberately. This means that the criticism that claims that TPB often neglects non-rational, and spontaneous decision factors is less prominent in this research context. Furthermore, the critique that TPB might not encompass all relevant factors, thus resulting in researchers often using extended versions of TPB makes sense in quantitative research, as surveys do not allow to delve deeper into the responses of participants. The semi-structured interviews used in this research allow participants to elaborate on answers, and if needed explain external factors they deem relevant to the subject at hand, such as economic conditions and regulatory policies that influenced their decision-making. The qualitative nature of this research allows for more flexibility and nuance upon questioning the different reasons behind the adoption of certain process innovations, therefore rendering the need for an extended version of the TPB unnecessary.

2.4 Conceptual Model

After having performed the literature review of existing barriers in both the restaurant and construction industry, the findings will now be briefly evaluated using the TPB framework.

2.4.1 Restaurant Industry

Figure 2 shows a conceptual model of how the different barriers affect the categories that influence people's intentions to adopt and implement process innovations in the restaurant industry.

Lack of trust in technology plays a role in shaping an attitude that negatively evaluates the adoption of process innovations such as automation, robotization, and digitization. Resistance to change of referent others such as employees or the customers shapes the subjective norm of the decision-maker. When employees or customers for example reject the implementation or adoption of a new process innovation it influences decision-makers in the restaurant business away from adoption. Lack of human capital, Infrastructural barriers, and financial costs all influence the perceived behavioral control (PBC). If companies perceive they do not have the required skills or financial resources necessary to adopt and implement a new process innovation this will likely result in the decision to not invest or implement a new process innovation. Infrastructural barriers function in the same way, if a restaurant owner perceives the implementation of robots as something that requires a large-scale renovation of the restaurant, it might decrease their confidence in the ability to guide that process innovation into a successful implementation. For the financial costs, if companies require additional funding because the

costs are too high this will likely decrease their sense of perceived behavioral control. On the other hand, PCB could be increased if the company knows or has experience with ways of finding additional sources of finance.

Literature review - barriers to process innovation adoption in the restaurant industry

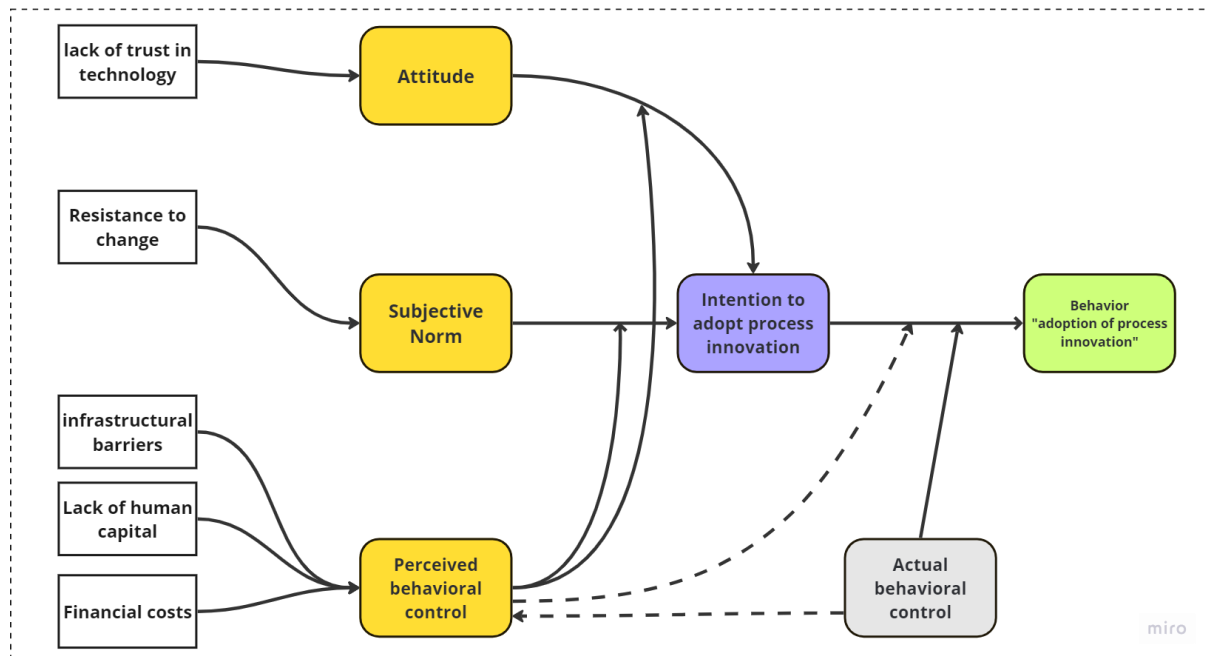


Figure 2 Conceptual model (based on the literature review) of the different barriers that affect the categories that influence people's intentions to adopt and implement process innovations in the restaurant industry.

2.4.2 Construction Industry

Figure 3 shows a conceptual model of how the different barriers affect the categories that influence people's intentions to adopt and implement process innovations in the construction industry.

A lack of technology awareness can influence attitudes, subjective norms, and perceived behavioral control. Concerning normative beliefs if fewer peer companies or competitors know about different process innovations, it becomes harder for companies to influence or to become influenced by each other. In addition, it also makes it harder for companies to learn about innovations. Lack of technology awareness can shape the attitude since having little knowledge of technology will likely create an attitude that is less interested in innovations thereby creating a self-reinforcing feedback loop.

Resistance to change in the form of employees resisting to work with a process innovation affects the subjective norm in the same way as it does for the restaurant sector. A lack of technology awareness, Lack of skills, and costs of implementation all influence the PBC because they affect the perceived resources and opportunities that a decision-maker has related to implementing process innovations. Firstly, regarding the lack of technology awareness, the decision-maker might not know about the possibilities for process innovations, this then bypasses his intention and makes it directly impossible for that person to adopt a process innovation. Secondly, if a decision-maker of a construction company perceives that he does not have employees with the right knowledge and skills available necessary for the implementation of the process innovation, it will force the decision toward not adopting it. The same holds for the cost of implementation, if the process innovation costs a substantial amount of money and

thereby threatens to eliminate or decrease the profit margins the decision-maker will be less inclined to adopt that process innovation.

Literature review - barriers to process innovation adoption in the construction industry

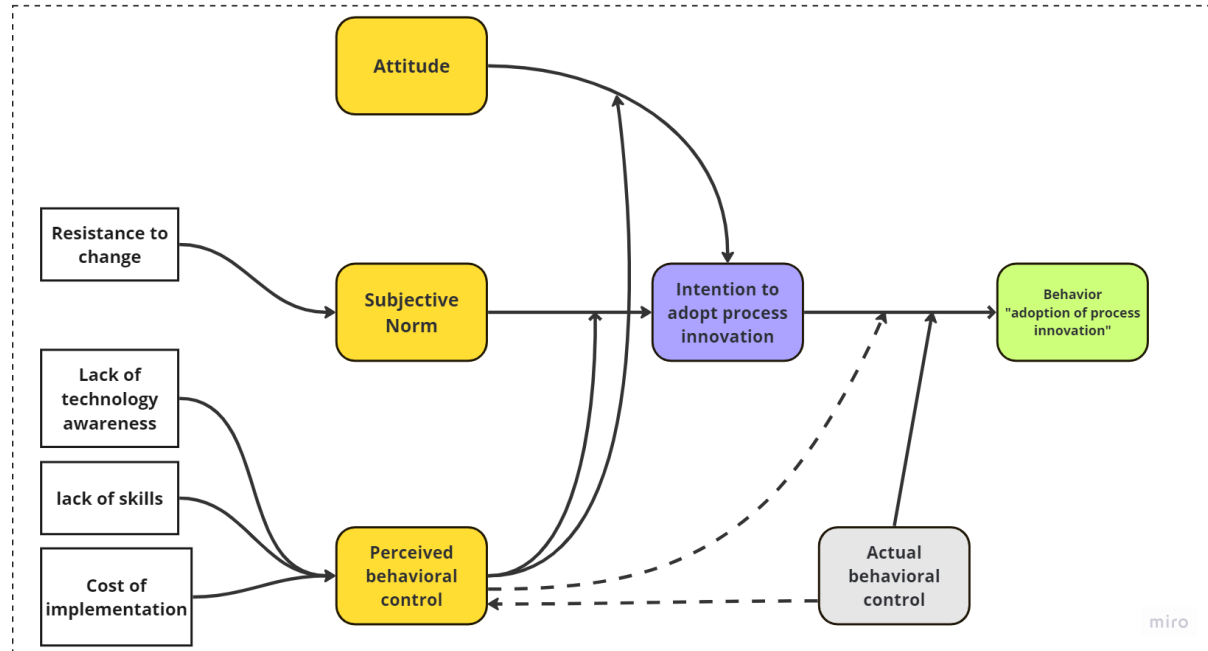


Figure 3 Conceptual model (based on the literature review) of how the different barriers affect the categories that influence people's intentions to adopt and implement process innovations in the Construction industry.

3. Methodology

3.1 Qualitative Research Design

This section explains the research approach that was used to answer the research question. This study used a qualitative research approach. The qualitative research approach was selected because while the general innovation trends within the selected sectors are relatively understood, the more specific motivations, experiences, and obstacles encountered by individuals in companies upon adopting particular process innovations remain unclear.

Qualitative research is suitable to address this gap since it intends to understand a particular social situation, event, role, group, or interaction (Creswell, 2009). The qualitative research approach allows for in-depth exploration and understanding of the underlying thought processes in the decision to adopt a process innovation or not, as well as understanding the experiences and evaluations of existing adopters. Furthermore, the qualitative research approach allows us to research the motivations and visions of the participants in a more flexible manner. This refers to for example, that in semi-structured interviews, the researchers can ask additional questions to issues that arise during the interview. This allows for the exploration of the underlying mechanism such as social, cultural, and situational factors that influence the adoption of process innovation.

This section continues by explaining the research steps that were performed for this study. The next section explains how the theoretical foundation was created for this thesis. The second section explains how participants were recruited for the study. The third section explains how the data was collected using semi-structured interviews. The fourth and final section highlights how the data was analyzed to gain the insights necessary to answer the research question.

3.2 Interviews

The interviews were conducted to gain insights to answer the sub-questions. As already mentioned, the interviews were based on the conceptual diagram (see Figures 2 and 3) that was created in the literature review. This means that the interview guide (Appendix A) is built of sections according to the factors of the Theory of Planned Behavior's (TPB) framework, referring to attitude, subjective norm, and perceived behavioral control. This was done to see which factors influence these categories and thereby the adoption of process innovations.

It is important to note that, since the participants share their personal perspectives on the adoption of the process innovation, this research will focus only on the perceived behavioral control and not the actual control. This is because actual behavioral control cannot be verified from the interviews alone. Fortunately, this is also in line with Ajzen (2019a) who stated that actual behavioral control is often difficult to determine and that perceived behavioral control “can serve as a proxy for actual control and be used for the prediction of behavior”. The section continues by discussing the sampling strategy, data collection, and how the data was analyzed.

3.2.1 Sampling Strategy

The sampling strategy used in this study was to contact restaurant owners and managers via LinkedIn. For the technical construction industry, referring to installation and construction companies, a wider range of job titles and professions were approached. Think of CEOs, innovation managers, project leaders, and lead designers. This was necessary because the value chains of these industries are more complex meaning that different input is required from different angles to capture the underlying mechanism that constitutes the adoption of process innovation. Before participants were approached, the researchers first looked into their job function and specialization. Again the online platform

LinkedIn was used to contact the different participants as well as the personal network of the researcher. In this research, the distinction between company size is made between large and small and medium-sized Enterprises (SMEs). The criteria used in this thesis are based on the criteria used by the Netherlands Chamber of Commerce (KVK and RVO, n.d.); fewer than 250 employees and an annual turnover equal to or less than €50 million. Companies with more employees and large annual turnovers are considered large. Table 1 provides an overview of this study's participants, detailing their job titles, the process innovation adopted at their respective companies, and the sizes of these companies. Note that some participants were also owners of the company.

Table 1 Overview of the participants in this study

Job title/ function	Type of business	Process innovations adopted	Size
<i>Restaurant Industry</i>			
1) Chief Financial Officer	Holding company different restaurant formulas	Process innovation depends on the restaurant's formula. Back-end systems are the same for all restaurants. QR codes	Large
2) Chief Operating Officer	Restaurant chain	Robot waiters Robots in the kitchen	Large
3) Restaurant owner	Restaurant	No process innovations	SME
4) Restaurant manager	Restaurant	QR codes	SME
5) Restaurant manager	Restaurant	Robot waiters QR codes	SME
6) 3D food printing and chef	Restaurant	AI recommendation bot 3D food printing machine QR codes	SME
<i>Construction Industry</i>			
7) Project manager	Construction contractor	BIM Prefabrication	Large
8) Business manager circularity	Engineering bureau with a focus on the building sector.	Sensor technology for predictive maintenance Data management AI	SME
9) Project developer	Construction developer	Integrated digital platform for business operations, e.g. including costs and planning	Large
10) Manager Production Optimization	Co-maker construction	BIM Prefabrication	Large
11) Chief Executive Officer	Designer and manufacturer of customizable solar installations	Digital application that allows clients to create designs ready for assembly.	SME
12) Architect/Account manager	Real estate support developer (wind, solar feasibility analysis)	BIM Grasshopper software to create parametric designs	SME
13) Strategy advisor sustainable housing	Construction and real estate company – Management and maintenance of buildings	Digitalizing information and dashboarding. Sensor technology.	Large
14) Manager business development	Construction and real estate company –	Digitalizing information and dashboarding.	Large

	management and maintenance of buildings	Sensor technology.	
15) Innovation process leader	Maintenance and improvement of existing homes	QR codes for personnel tracking on site. Application for document sharing among project partners.	SME
16) Strategy Designer	Construction robotics	Robotic bricklayer	SME
17) Director-major shareholder (DGA)	Total installation company (“Totaalinstallateur”)	BIM Prefabrication	SME
18) Director-major shareholder (DGA)	Roofer company (for renovation and new construction)	Aerial software application for offsite roof inspection Drones	SME
19) Director	Designer, constructor, and maintainer of electrical installations in the building sector.	BIM Prefabrication	Large
20) Sales director	3D modeling & simulation company for a variety of sectors including Architecture, Engineering, and Construction	3D simulation and scenario modelling	Large
21) Director-owner	Modular and circular housing assembler	Semi-automated assembly line factory BIM Prefabrication	SME

3.2.2 Data Collection

For the collection of data, face-to-face in-depth semi-structured interviews were used. The interviews were conducted from 15-02-2024 to 28-03-2024. The interviews were mainly conducted online using the Microsoft Teams software. The interviews lasted an average time of 52 minutes per participant. Most interviews lasted in the range of 40-65 minutes, however, there was one notably shorter and one notably longer interview. The shortest interview (23 minutes) was conducted with a restaurant owner who had not implemented any process innovations and was concise in the explanation of his reasons. The longest interview was conducted with a construction project developer, who explained first the digitalization of the internal processes of the company he worked for. In addition, he explained the interactions with contractors and architects occurring in the realization of a project, in reaction to getting asked how innovations in the construction of the building occurred. Before the interview, the participants received an informed consent form which they were asked to sign. The interviews were recorded and transcribed. The interview started with a short introduction to the topic by the researcher. The interview consisted of a set of open and closed questions. The questions were categorized using the TPB framework and provided information regarding the participant’s attitude, subjective norms, perceived behavioral control, and intentions toward adopting respective process innovations. The same questions were asked to existing adopters of different process innovations, to gain insights into their attitudes, subjective norms, and perceived behavioral control.

3.2.3 Data Analysis

First, the researcher created verbatim transcripts of the interviews. This means that identifiers were removed from the data to protect the participants’ anonymity. The analysis of the verbatim transcripts focused on the informational content, which also contains the social and cultural meanings attached to

this content. Afterward, a thematic analysis was performed using the NVivo software tool. The transcripts were first coded. Coding refers to assigning words or short phrases to ideas, and issues that appear in the transcription data (Hennink et al., 2020). Here both an inductive and deductive approach to coding was used. The deductive approach means that codes found during the performed literature study were used to structure the topic of the interview guide, and also to validate whether the empirical data reflects the barriers to adoption found in the literature review. The inductive approach was used to derive new insights from the empirical data, that is codes were derived directly from the transcript data. After the development of the codes, the categorization and conceptualization of the codes were done. Categorization involves assembling a set of codes that collectively embody a larger concept (Hennink et al., 2020, p. 225 – 262). These categories or themes are grouped into an overall thematic map, which will aim to give an explanation or tell a story of how attitudes, subjective norms, and perceived behavioral control influence the intention and adoption of process innovation.

4. Analysis Results

This section has the main aim of explaining which barriers and drivers influence the decision to implement process innovation among companies in the restaurant industry and the construction industry. To answer the main research question, we first need to understand which driving factors shape the attitude, subjective norms, and PBC of the participants toward the adoption of process innovations. Similarly, we also need to understand which barriers or obstructing factors shape the attitude, subjective norms, and perceived behavioral control away from adopting process innovations.

It is important to note that the barriers and how they shape the different variables that lead to the adoption of process innovation are discussed individually for clarity purposes, in the sense that the reader gets an idea of how the individual factors can influence the specific category of TPB. In reality, the obstructing factors interact and depend on the situations at hand. The goal is therefore to get an understanding and overview of the factors that play a role in shaping the categories into adopting process innovations, and the opposite of shaping the factors away from process innovations.

The framework as depicted in Figure 4 will serve as the basis of the analysis from which the following questions are answered:

- What are the barriers or obstructing factors that shape the attitude, subjective norms, and perceived behavioral control toward not adopting process innovations? (SQ1)
- What are the driving mechanisms that shape the attitude, subjective norms, and perceived behavioral control toward the adoption of process innovations? (SQ2)

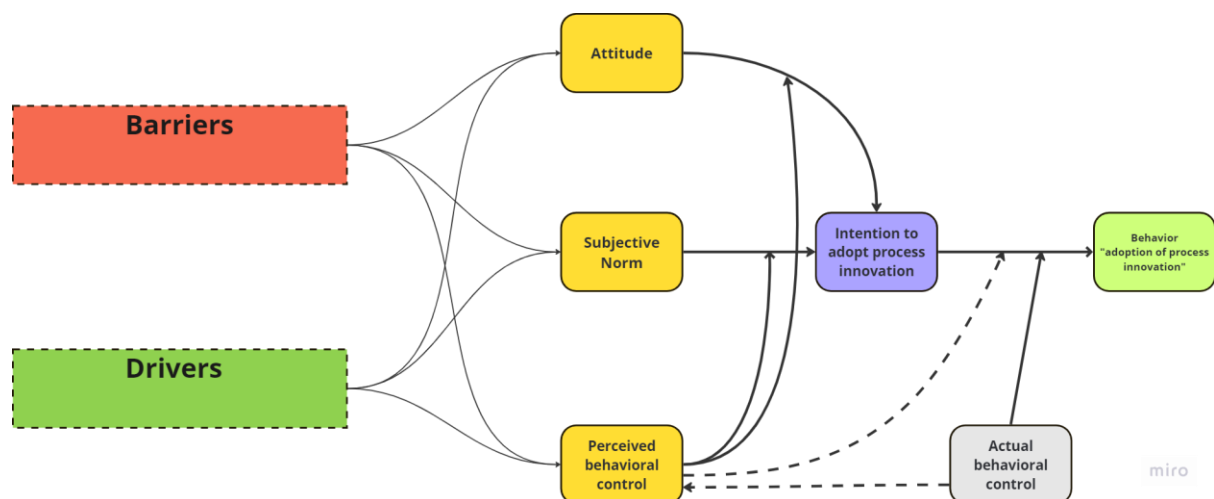


Figure 4 A schematic overview of the Theory of Planned Behavior for the adoption of process innovation

4.1 Barriers to Process Innovation in the Restaurant Industry

This section will start by investigating which obstructing factors shape the attitude, subjective norm, and perceived behavioral control (PBC) of decision-makers in restaurants away from the adoption of process innovations. Figure 5 shows a conceptual overview of the barriers in the restaurant industry. The individual barriers and how they influence the respective category of the TPB will be explained in detail.

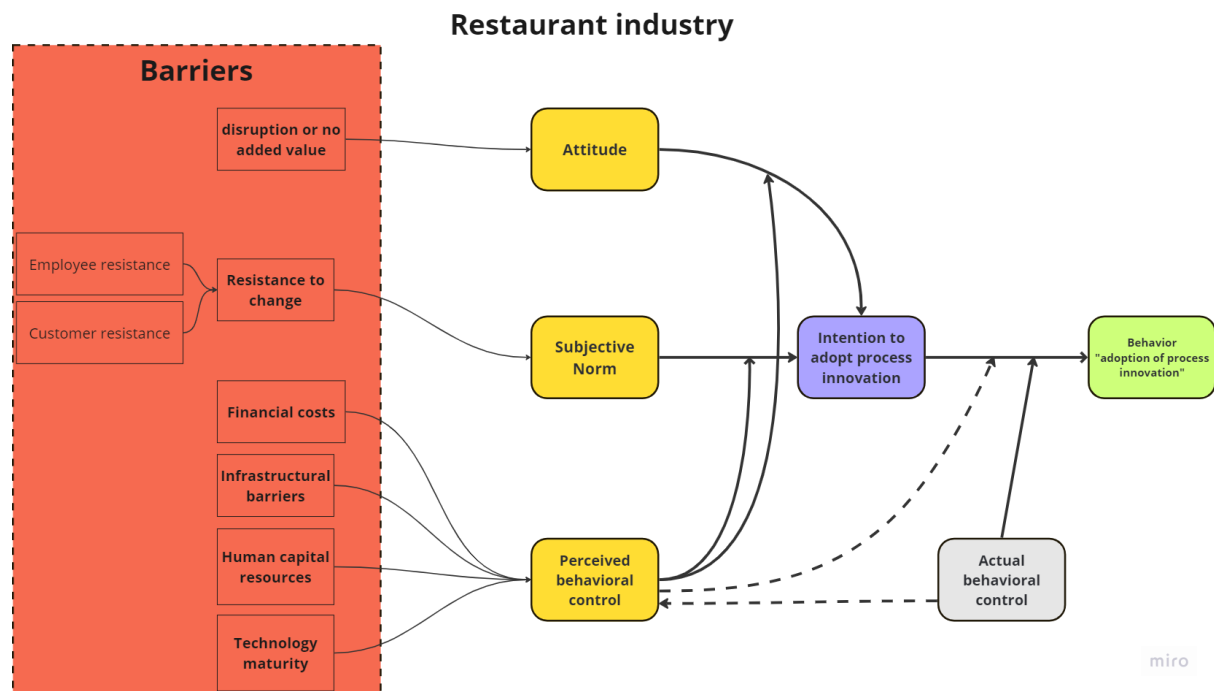


Figure 5 A schematic overview of the barriers in the Restaurant industry

4.1.1 Attitudes – Barriers Restaurant Industry

Innovation disrupts or does not influence the value proposition

This section focuses on the behavioral belief factors that can act as a barrier and shape negative attitudes towards adopting and implementing process innovations, thereby disrupting the adoption. It is important to repeat here that according to the TPB attitudes toward a behavior, are preceded by the behavioral beliefs that link the behavior to an expected outcome or experience (Ajzen, 2019a). This means that a behavioral belief that links the adoption of a process innovation to an outcome that we value as unfavorable, the attitude toward the adoption of that process innovation will be labeled as negative.

A clear example of such a behavioral belief was found in Interview 4, where the restaurant manager defined a desired outcome or goal that he wanted to deliver with his restaurant. The goal is to create hospitality value for the customer. According to him, this is done by creating and maintaining a loyal circle of customers (Interview 4). This expresses itself for instance when the customer comes into the restaurant and you already know what they want to order. So it is really about personal contact with regulars “and I do not think you can achieve that by using robots” (Interview 4).

In this example, the manager has the behavioral belief that the implementation of robots in the restaurant disrupts or does not facilitate the desired outcome of creating and maintaining a loyal circle of customers. It then makes sense that he is not willing to adopt for example waiter robots that help in the restaurant.

Another way attitude can form a barrier to the adoption of process innovations is described by the restaurant owner in Interview 3. He discusses how an important factor for the adoption of process innovation lies in the size of the company. He owns a single restaurant establishment and has no trouble for example doing the bookkeeping by hand, and therefore feels no need to buy or lease a digital application (Interview 3). He describes that, in his behavioral belief the implementation of a process innovation, QR codes or robots in the kitchen or food serving, has no significant influence on the outcome, that being the service that the restaurant delivers to the customer. Then again there is no reason why he would adopt a process innovation.

Trust

The trust barrier was identified in the literature, referring to the lack of trust between technology and humans, especially trust between robots and humans. This barrier was however not found in this research. A reason the participants in this study did not mention this barrier, could be because the literature described the barrier to be of concern to the customer. Grobbelaar et al. (2021) explained that customer's lack of trust comes from the questioning of the robot's ability to follow strict safety standards. The decision-makers in this study might have not heard about customers complaining about this issue concerning their restaurant robots.

4.1.2 Subjective Norms – Barriers Restaurant Industry

Consumer and employee resistance to change

This section looks at which factors influence the subjective norms of restaurant decision-makers away from adopting process innovations. Subjective norms are determined by the underlying normative beliefs which are concerned with the likelihood that important referent individuals or groups approve or disapprove of performing a given behavior (Ajzen, 1991). So in the context of adopting process innovations, referent others can be related to whether employees and customers approve or disapprove of the adoption.

Both can pose a barrier to the adoption of different process innovations. In interview 6 for example, the general perception of people on AI and robots was mentioned as a barrier to adoption. The 3D food printing chef works in an innovative restaurant, that tries to implement a variety of process innovations (see Table 3), with the main purpose of utilizing these technologies so that they can maximize the personalized value to the customer. But they have received reactions from customers that “it really scares some people off. We actually hear that from customers. That they come in here and are surprised that there are no drones flying around and robots driving around” (Interview 6). What is interesting about this quote is that people have a certain perception and expectation of what an innovative restaurant that states the use of AI on their website will look like. This perception, that of drones flying around, can form a barrier for them to decide to go to the restaurant.

On the employee side, this problem can also occur, when the COO of a restaurant chain (Interview 2) told their employees that they started with the implementation of robots their first reaction was that of fear of job loss, “Will I lose my job?” or “are you not going to need anyone anymore?” (Interview 2).

Another problem that can arise is that employees have difficulty working with or cooperating with the new process innovation. A restaurant manager describes how the implementation of QR codes posed a challenge at first because the employees had difficulty interacting with the customers as they could no longer walk to a table and ask “Do you want something to drink” (Interview 4). This then resulted in a more passive behavior of the waiters, in which they sometimes forgot to say simple things to guests such as “enjoy your meal” (Interview 4). This eventually also led to a more balanced approach between the QR codes and personal ordering. Instead of only relying on the QR codes.

4.1.3 Perceived Behavioral Control – Barriers Restaurant Industry

Next to the belief and perception factors discussed above, some factors directly influence an individual's capabilities to perform the behavior, in this case, the adoption of process innovations. These factors are the required resources and opportunities needed to perform the behavior, termed under the perceived behavioral control (PBC) factor (Ajzen, 1991).

Financial costs

Financial costs can pose a resource barrier as was also identified in the literature. This barrier was confirmed in the conducted interviews. The barrier refers to that the implementation of process innovation usually requires high upfront costs, which almost always plays a role in the decision to implement any kind of process innovation (Interviews 1, 4, 5, 6).

Interestingly, interview 1 added a factor to this barrier, in that the restaurant sector as a whole has endured tough times because of the COVID-19 pandemic. A lot of restaurants had to secure loans to survive this period. This means that a lot of restaurants are working on paying back the debts on these loans. So there are “relatively few liquid assets in the sector to perform these type [refers to different process innovations] of investments” (Interview 1).

Infrastructural barriers

This barrier was also identified in the literature review. Multiple aspects of this barrier were also found in the conducted interviews. First of all, the physical space in the restaurant required for robot implementations was indeed mentioned by a restaurant manager (Interview 5). He discussed that the space between tables plays an important role in the implementation of robot waiters because you “need to have spacious, wide aisles, and wide openings leading to dining halls” (Interview 5).

The second aspect of this barrier was that when restaurants have multiple establishments, or they work under a larger holding company, then the process innovations have to be scalable in the sense that they can be rolled out over multiple establishments. This requirement results in that many innovations are no longer suitable for implementation (COO, Interview 2). The COO mentions the example of a new software program that can be used to make smarter and optimized planning for a single restaurant but is not ready to be implemented across different establishments. So it is important for restaurants that are part of chains that process innovations are scalable.

A final aspect of the infrastructural barrier was mentioned by the CFO of the restaurant holding company (interview 1) and is related to the use of automation in the kitchen. Robotic automation requires the use of substantial electricity, and most of the grid connections that their restaurants have are simply too small for the required energy necessary for the automation. The Netherlands is currently dealing with grid capacity problems (van Gastel, 2023), meaning that it will be difficult to upgrade grid connections, making it hard to invest in machinery in the kitchen. “We currently have multiple locations where we do not have enough electricity, resulting in that we cannot for example install a fourth pizza oven” (Interview 1).

Human capital resources

This barrier was also identified in the literature review and refers to the lack of technical knowledge related to technologies that are implemented in different process innovations. It is indeed true that for all the different innovations that were mentioned none of the process innovations were developed by in-house employees, from the implementation of an AI recommendation robot (Interview 6), to the use

of robots for serving dishes (Interviews 2 and 5). The development and implementation of these innovations were always done in cooperation with a third-party supplier.

For the implementation of robot waiters, the restaurant collaborated with a supplier of the robots (Interview 2 and 5), the same holds for the implementation of QR codes (Interview 4). However, outsourcing the technical side of the process innovation was never mentioned as a barrier, and neither did the restaurant owners expect the employees to have technical knowledge regarding the process innovation, other than that they were trained and required to learn to work with and operate the technology to integrate it into the existing business processes. Although problems occur in this interaction with the technology as discussed in the subjective norm barrier section, decision-makers do not mention that their employees require a certain knowledge level or skills regarding digitalization and automation. Instead, the decision-makers have a clear demand for every process innovation, as expressed by the COO (interview 2) “In principle, everyone should be able to master the new implementation within 10 to 15 minutes, otherwise, I don’t consider it a good innovation and I won’t be able to implement it”.

Technology maturity

Another important barrier that was mentioned by both adopters and non-adopters within the restaurant industry regarding for example robot implementation was the perception that current robotic technology is not mature enough to the point that it can match or replace a human worker, particularly in terms of achieving the necessary top speeds during peak hours (CFO, Interview 1).

A similar remark was made by the 3D printing chef of an innovative restaurant company that started with the implementation of AI. The restaurant used an AI recommendation chatbot that would incorporate information about allergies and preferences. But this AI chatbot is still in development and “after two months of training, when somebody for example said they had a shellfish allergy, they [the customer] would still get a shrimp dish recommended, which directly contains those allergens” (Interview 6). To solve this issue, employees had to help customers out frequently. Eventually, this resulted in the restaurant deciding to temporarily stop the implementation of the AI Chatbot, although there is still a launching roadmap in place.

4.2 Drivers to Process Innovation in the Restaurant Industry

This section discusses the driving factors that influence the attitude, subjective norms, and PBC of restaurant decision-makers toward the adoption of process innovation. A conceptual overview of the drivers can be seen in Figure 6.

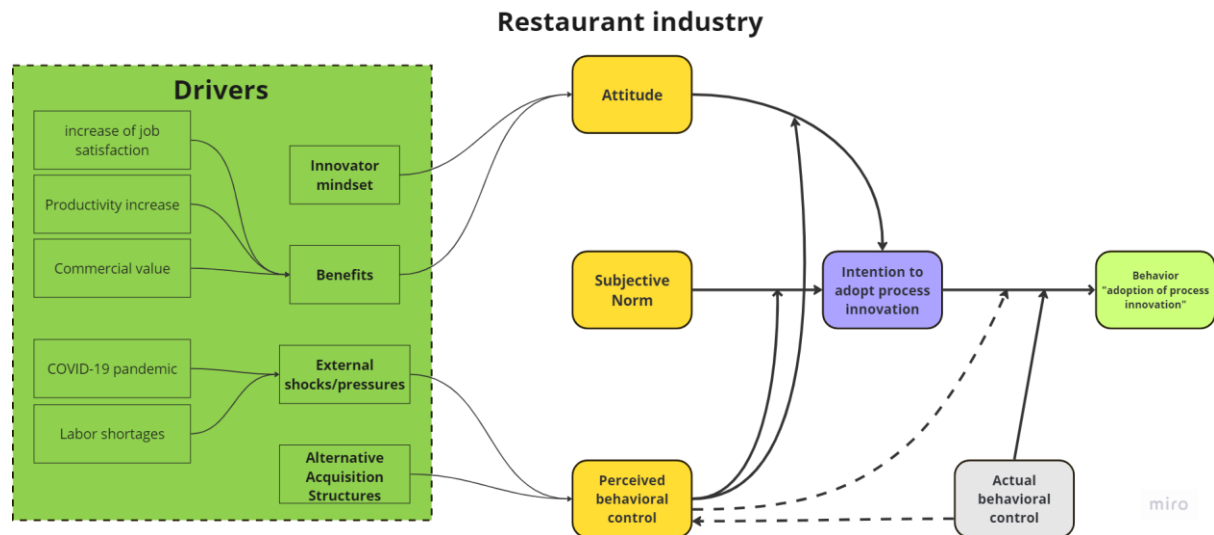


Figure 6 A schematic overview of drivers in the Restaurant sector

4.2.1 Attitudes – Drivers Restaurant Industry

In this section, we identify how behavioral beliefs can shape attitudes that increase the likelihood of process innovation adoption. Starting with the clear mindset difference between an innovator and a non-adopter

Innovator Mindset

From the interviews, it becomes clear that individuals who focus on adopting process innovation, perceive the implementation of process innovation as positive, which originates not directly from the process innovation as an entity itself, but rather from the relation it has to their goals or their evaluation of the outcome. Most restaurant owners and managers are quite clear about the main goal that they have. Look for example at the description that the COO (Interview 2) of a restaurant chain that innovated with robotization stated about the value creation of his restaurant. “I do think it is very important that you apply innovation not for the sake of applying innovation, but to improve things for the customer, but also for your own employees. And of course, ultimately this is also good for your own wallet” (COO, Interview 2).

Innovators and early adopters in the restaurant sector for example value the adoption of process innovation as a means to an end. The end is the value creation for the customer, in most cases the hospitality value for the customer. Whether or not to adopt process innovation, is dependent on how it relates to this goal. That is, does the individual perceive (behavioral belief) that the process innovation can enhance the customer experience of their restaurant? In that case, a positive attitude is shaped towards the adoption and implementation of the process innovation, since innovation functions as an enabler for the final goal (outcome evaluation). “We want to serve good food for an affordable price, and we can only do that by implementing robots, because it enables us to require fewer workers, so we

can keep the costs low and also the price to the customer. So you see, the robots actually help us to realize our strategy” (COO, Interview 2).

And this is exactly how the attitude can be shaped both positively and negatively toward the adoption of process innovation. Restaurant owners and managers with a negative attitude toward process innovation describe the innovations as disruptive to their goal of creating hospitality value, whereas innovators see technology as an enabling entity for their overall goal.

Look for example how attitude can play a role in the face of a financial barrier. To illustrate how attitude shapes the perception of this barrier, it is important to note that the innovator is focused on how the process innovation enables them to achieve their desired outcome goal, rather than focusing on the barrier that obstructs the implementation. This means that when one cannot finance the process innovation at the moment, “you can go to a bank, you just go to fix that, but there are also so many subsidies and these type of things through which you can make it work.” (COO, Interview 2). But this comes with a different mindset and effort, and is also recognized by the COO himself “You need to be knowledgeable about it, and you need to be interested in it. You must, you have to be willing to ride that rollercoaster, because your innovation does not work from day one” (Interview 2).

Benefits

There are several benefits mentioned by the participants, upon implementing different process innovations.

1. Robots decrease (physical) workload and thereby increase job satisfaction

The first benefit of the implementation of for example the implementation of robots in the existing work processes of restaurants is that the current labor shortages also result in a higher workload for existing workers. This increased workload can cause some workers to quit their jobs (Interview 5). Robots can support the workers in delivering dishes to the customer, as well as support the workers in moving dishes away from the tables. This results in workers having to perform less walking and carrying of dishes themselves, which decreases the workload and increases job satisfaction (Interview 2 and 5). “And if you can lower the workload for your employees by using robots, then you also increase the chance of them staying” (Interview 5).

2. Productivity increase

Different process innovations can increase the productivity of the restaurant in various ways. For example, a restaurant manager (interview 4) explains how before the QR codes, the restaurant had district runners that would be responsible for overseeing certain “districts” of the restaurant. Their responsibilities encompassed that customers’ needs were fulfilled, regarding food and drinks. However, with the implementation of the QR codes guests can manage their orders for food and drinks, rendering the district runner role obsolete. Attention now shifts to cleaning tables and laying out cutlery for new tables. This responsibility has been divided among existing staff, such as the food runners and drink serves, thereby being able to do the same work with fewer people effectively increasing productivity (Interview 4).

The incorporation of robots into the existing work processes can also increase the productivity of the restaurant. In interview 5, the restaurant manager explains that the waiter robot helps in decreasing the number of steps the workers have to take. He mentions that when clearing tables, one can load the entire robot with empty plates and glasses. In addition to serving food, the robot can bring three layers of trays with food, resulting in three people not having to walk back and forth between the kitchen and the customer. Especially during busy periods, it reduces the amount of walking for existing staff (Interview 5). This benefit was also confirmed by the COO of a different restaurant chain, who stated that the

decrease of walking distance for staff, led to the need for less personnel (Interview 2). In addition, because not every restaurant in the chain uses the same amount of robots, because of reasons explained by the infrastructural barrier, the labor costs are higher in restaurants with less, again stressing the productivity value of the robots in the restaurants (Interview 2).

3. Innovations can be used as commercial value and selling point

An additional benefit of the use of different process innovations it be used as a unique selling point to the customer. In Interview 5, the restaurant manager explains how children are particularly interested in the waiter robots, and that they get excited about it. This then results in parents often coming to the restaurant “purely because the children like it” (Interview 5). The restaurant owner in Interview 3 did not adopt any process innovations himself and was not interested in using robots as waiters. However, he did explain how his nephew opened a restaurant that used waiter robots and also heavily promoted that as a selling point. “But yeah, it was more for the show than for actual productivity” (Interview 3). Other restaurants recognize that “the customer also likes it”, but that is taken more as an additional benefit, rather than as the aim of the implementation (Interview 2).

For the restaurant that implemented multiple process innovations, the different technologies are used for commercial value. The Food chef highlights how embracing innovation was used as a strategy to “really stand out” (Interview 6).

4.2.2 Subjective Norms – Drivers Restaurant Industry

There are not a lot of driver mechanisms in place from referent others that stimulate the adoption of process innovation. Restaurants do look at other restaurants for inspiration. The CFO of the holding company for example discussed how MacDonald from an optimization standpoint is doing great (Interview 1). but they do not experience pressure to adopt similar practices.

It is, however, interesting to add that where the disapproval of employees toward process innovation can be a barrier to the adoption of process innovation in restaurants. The opposite could also happen, where employees who experienced the workload reduction because of robots, now rather work in restaurants that have robots (Interview 5). In this way, robots can ironically be used as a tool to keep employees from leaving.

4.2.3 Perceived Behavioral Control – Drivers Restaurant Industry

External shocks

Upon looking at driving factors that influence the PBC of decision-makers, it becomes evident that these factors create some form of necessity, that in a way can override existing attitudes and subjective norms that would not favor the adoption of a process innovation. Necessity here refers to that the restaurants usually face some form of external pressure or shocks that forces their hand to act.

1. COVID-19 pandemic

A clear example is that of the COVID pandemic that has hit the restaurant sector notoriously hard. In Interview 4, the manager discusses how the COVID-19 pandemic forced the restaurant to downsize its workforce because of the safety prescriptions. This problem was then solved by implementing the QR code (Interview 4). Now that the pandemic is over, the restaurant still uses the QR codes. Thus this external shock forced the restaurant to innovate their processes by adopting a digital solution.

A similar situation was described by the CFO of the restaurant holding company, which explained the COVID-19 pandemic as the driver for their integrated concept approach, which hosts different restaurant formulas all under the same company (Interview 1). The restaurants look different and serve different foods and drinks, but all the restaurants run on the same operating systems, such as the payment system, so all the restaurants share the same back office. This also allows them to test different process innovations in different restaurants and locations. If an innovation is deemed successful, it can be rolled out in potentially 73 restaurants (Interview 1).

2. Labor shortages

Another form of external shock that can act as a driver for the adoption of different process innovations is labor shortages. However, the opinions among the participants vary.

For larger companies solutions are found in implementing different process innovations (Interviews 1 and 2). The type of worker that is hard to find or that the process innovation focuses on differs per company. The CFO in Interview 1 explains that most of the shortages they experience are related to kitchen personnel. This also means that the team that is responsible for process optimizations, focuses mainly on process innovations that can save labor in the kitchen (Interview 1).

For smaller companies, the answers were different. Interviewees did not see the implementation of process innovations such as robots as a way to deal with labor shortages (Interview 3, 4, 5, 6). Solutions are often found in different actions, for example, the manager in Interview 4 mentions that he would rather close certain parts of the restaurants and “limit the range of products and the number of visitors” (Interview 4). The chef in Interview 6 did mention that process innovations could help in the service side of the restaurant because that was originally their vision with the use of Artificial Intelligence (Interview 6). However, he also mentioned that it was harder to find workers in the kitchen and that automating kitchen operations, for example by using a robot is not something they have investigated. After asking why, he responded “Because it [robots] does not fit in our philosophy, we want to stay away from robots to keep delivering the human experience” (Interview 6).

An important addition to the notion of the relationship between labor shortages and process innovation is that, yes process innovation can help with reducing labor (Interview 2, 6), but an important focus point should also be keeping the jobs that are necessary in the restaurant industry alive (Interview 1, 2). The COO in Interview 2 for example states “Yes innovation is good, is important. But let’s also go back to the basics. How do we keep people in the industry?” (Interview 2). A similar notion was made by the CFO of interview 1, who stated that we need to start investing in this profession. He then continues that they as a company are part of initiatives that show that people can make a career in the hospitality sector. The goal of this is “that the profession will also continue to thrive among the next generation” (Interview 2).

Alternative Acquisition Structures

As some restaurants cannot pay for the high upfront costs of most process innovations, suppliers of technologies have come up with alternative strategies to sell their products. The restaurant manager in Interview 5 for example discusses that a supplier of waiter robots approached a large set of restaurants and gave them the choice to test their robots. When they accepted the supplier's offer, it was possible to gain experience and use the robot for several months free of charge. This approach decreases the perceived difficulty and improves the perceived ease of adoption for the decision-maker since it now requires fewer resources to adopt the robot. The result was that “if you have already tested it, and because yeah after a few months, you have kind of already implemented it in your system, so then it becomes hard to say no”. (Interview 5). Note that the TPB can only explain the behavior in a particular moment, so it could be that during the free user period, the attitudes toward the robots positively change.

In an alternative approach, the supplier of waiter robots engaged in a collaboration with a restaurant that had gained a name for innovating with waiter robots. The COO (interview 2) explains how a collaboration between the restaurant and the robot supplier created benefits for both sides. The restaurant was allowed to use and experiment with innovative robots, and the supplier gained hands-on and real-life experience in a new European market at the time. The collaboration also allowed for a quick feedback loop for the development of the robots. When the restaurant had a certain feature request or software/hardware bug, then the supplier would incorporate that feature, or solve the bug and a week later it was incorporated and operational (Interview 2). This sped up the development of the robots substantially and thus both sides gained from the collaboration.

4.3 Barriers to Process Innovation in the Construction Industry

This section discusses the obstructing factors that influence the attitude, subjective norms, and PBC of construction industry decision-makers away from the adoption of process innovation. A conceptual overview of the barriers in the construction industry can be seen in Figure 7.

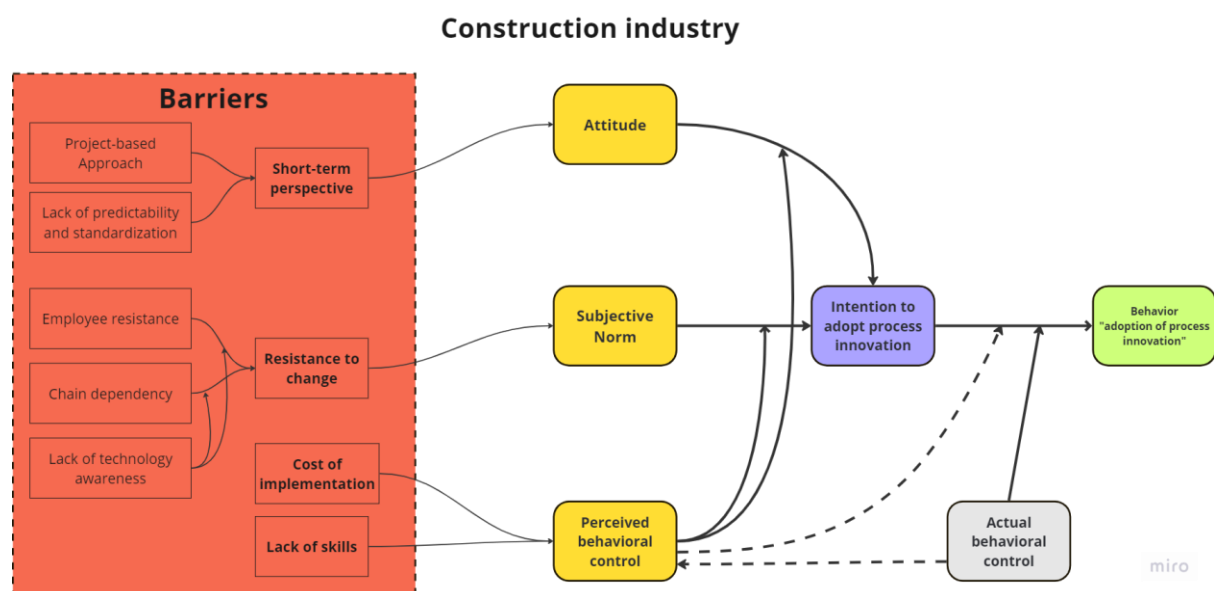


Figure 7 A schematic overview of barriers in the Construction industry

4.3.1 Attitudes – Barriers Construction Industry

Short-term perspective

In the implementation of innovations, hurdles are eminent to occur. Implementing innovations takes time and often requires some trial and error. It is not surprising that one of the main attitude barrier themes that arrived during the execution of the interviews is the construction industry's focus on the short term. This short-term perspective reveals itself as a common theme and expresses itself into several subset barriers.

Note that the short-term focus on profit is a barrier to the adoption of process innovations, however this will not be discussed here, but rather under the PBC section 4.4.3 on *Cost of implementation*.

To understand how this short-term perspective creates a barrier to the adoption of different process innovations, it is first important to explain how the short-term perspective is embedded in a project-

based approach and how this creates a lack of predictability for the different parties involved in the construction chain. It will become clear how this lack of predictability makes it hard to adopt process innovations.

Project-based approach

One of the ways that the short-term perspective can become a barrier to the adoption of process innovations is because of how it is embedded in the project-based approach. This approach can best be explained by the following quote from a project manager who works for a large contractor company, “we always think in, one of a kind, we always make a unique product or always a unique building. We are used to thinking in project-based solutions instead of process-based or project transcending [solutions]” (Interview 7).

This project-based approach was mentioned in a variety of interviews (Interviews 7, 9, 10, 11, 18), and seems to form a barrier to the adoption and implementation of process innovations in different subtle ways. As the majority of the construction chain is used to working in this project-based method of working, the demand changes for every project. When no project is the same, no building is the same and thus clients can ask for a different type of building each time.

Before continuing, it is important to point out that the project-based approach is evaluated purely based on its relation to the adoption and implementation of process innovations. This means that it is not evaluated based on other aspects such as the aesthetic aspects.

Lack of predictability and standardization

Because no project is the same, the contractors in the construction chain have to find project-specific solutions, which in and of itself is not a bad thing. However, in the situation that a company might want to increase the productivity of its processes through the adoption of digitalization and automation, a problem arises. This problem arises because digitalization and automation are intimately linked to some form of standardization, as you need a certain amount of predictability in your processes to digitalize and automate them. It is difficult to find predictability when every project could be different. This is how the project-based approach creates a sense of unpredictability among the parties in the construction chain that are responsible for realizing the project.

To get a more in-depth understanding of how this creates a barrier to the adoption of process innovations, we turn to the explanation of a housing production manufacturer (Interview 10). On getting asked which factors he perceived as hindering the implementation of process innovations, the answer was “Yeah, just the changing demands of the customer, that’s just really, that’s just really killing” (Interview 10). The company is trying to build a factory in which they can produce houses efficiently using both automation and digitalization. However, it is difficult to design this factory and the processes within it when the houses that it is supposed to produce change for each project.

Another dimension through which the lack of predictability is created by changing demands is through government legislation. This is illustrated by the sales director (Interview 20) of a multinational company that does 3D modeling and simulation for a variety of sectors including the construction sector. He outlines the example of an entrepreneur that for example is considering adopting a new digital tool in reaction to government legislation (see section 4.4.2 Demand-based) but needs to get a loan to finance the implementation of the digital tool. If the government changes the policy-making every 3 to 4 years, for example, because of elections, then the newly implemented process innovation might be rendered obsolete, meaning that the entrepreneur sees no return on investment and cannot pay back the loan on the investment, and then you have a problem (Sales Director, Interview 20).

Ajzen (1991) explained how attitudes link the behavior to a certain outcome, and that people thus come to favor those behaviors they believe have the most desirable consequences. The behavior in this case is the adoption and implementation of process innovations. In the project-based approach or the short-term perspective, the adoption of process innovations is linked to the outcome of successfully delivering projects since this creates the revenue necessary for the company's existence. When the demand changes for every project, the adoption of process innovation which requires a form of standardization, is labeled as negative or obstructing to the desired outcome, and therefore forms a barrier to the adoption and implementation of process innovations. This then explains why "in construction, R&D and innovation are often labeled as obstructing the process, rather than something desirable, because often everyone tends to think short-term. I just need to order my piles tomorrow, why are you now burdening me with some new program?" (Interview 10).

4.3.2 Subjective Norms – Barriers Construction Industry

Resistance to change

As discussed in the literature review, the construction sector is not known for its ability to embrace change (Demirkesen and Tezel, 2022). This barrier was also identified in the conducted interviews. In the context of adopting process innovations, decision-makers in the construction industry can face resistance to change from two referent other groups. The first group regards the internal forces, encompassing the resistance from employees of the company. The second refers to the external forces on a company, originating from the chain-like structure inherent in the construction sector.

Employee resistance

The first example of referent others comes in the form of employees that resist change or resist the process innovations that cause the changes. This has been mentioned throughout different interviews (interviews 7, 13, 14, 17). The reasons for their resistance can vary. The Director of a family construction company (Interview 17) explains that he interprets resistance from employees as a signal, as the "primary behavior", and he tries to understand the underlying cause of the resistance. Sometimes people find it difficult and do not understand the new process innovations and they need more time to adapt (Interview 17).

Similarly, workers resist the implementation with seemingly no rational reasons, because the workers for example, do not understand the benefits of the process innovation (see lack of technology awareness 4.3.3). A business developer for a large contractor company explains how during a digitalization initiative, mechanics transitioned from paper-based task assignment and reporting to utilizing digital tablets. Feedback during training sessions included comments such as "the tablets are too small and it is so complicated", yet during the coffee breaks, the mechanics were consistently typing on their substantially smaller smartphones (Interview 14). Afterward, he noticed that it had more to do with that the mechanics did not understand the benefits of the digital tablets and saw no initial use for their implementation.

Resistance from employees can become a significant barrier to the adoption of process innovation, especially in larger organizations. A strategy advisor for sustainable housing (Interview 13) explains how a change manager had to convince an astonishing diversity of decision-makers in the company, some not even directly related to the innovation project. "But these are then people who oversee certain funds or budgets" (Interview 13). In addition, the board can interpret the resistance of employees as a sign to stop the innovation implementation and focus more on the stability of the company (budget-cutting 4.3.3). This disrupts the speed and adaptability you need to succeed in innovation projects.

Chain dependency

The second source of referent others that can form a resistance to the adoption of process innovation originates in the architecture of the construction sector. The construction sector is made up of a chain-like structure, in which a chain of individual sub-contractor companies is often involved in the realization of a project (Interview 9). Before discussing how this chain dependency can form a resistance to the adoption of process innovation, a short elaboration is given about the architecture of the construction sector.

The project developer (Interview 9) explained that in the construction sector, a client outlines the requirements for a specific project along with a purchase price. The project developer is responsible for realizing the project within the specified budget, while also ensuring their profitability. Collaboration occurs between the project developer, the main contractor, and sometimes external parties such as architects bureaus. In this collaboration, the parties determine how the project will be realized. The main contractor “knows the market” (Interview 9), meaning that the main contractor knows the different suppliers and subcontractors, sometimes referred to as co-makers (Interview 7, 9, 21) that could be engaged in the project to realize it. The chain-like structure was illustrated by Interview 9 and confirmed by others (Interview 7, 8, 10, 11, 14, 15, 17, 18, 20, 21), a schematic overview can be seen in Figure 8.

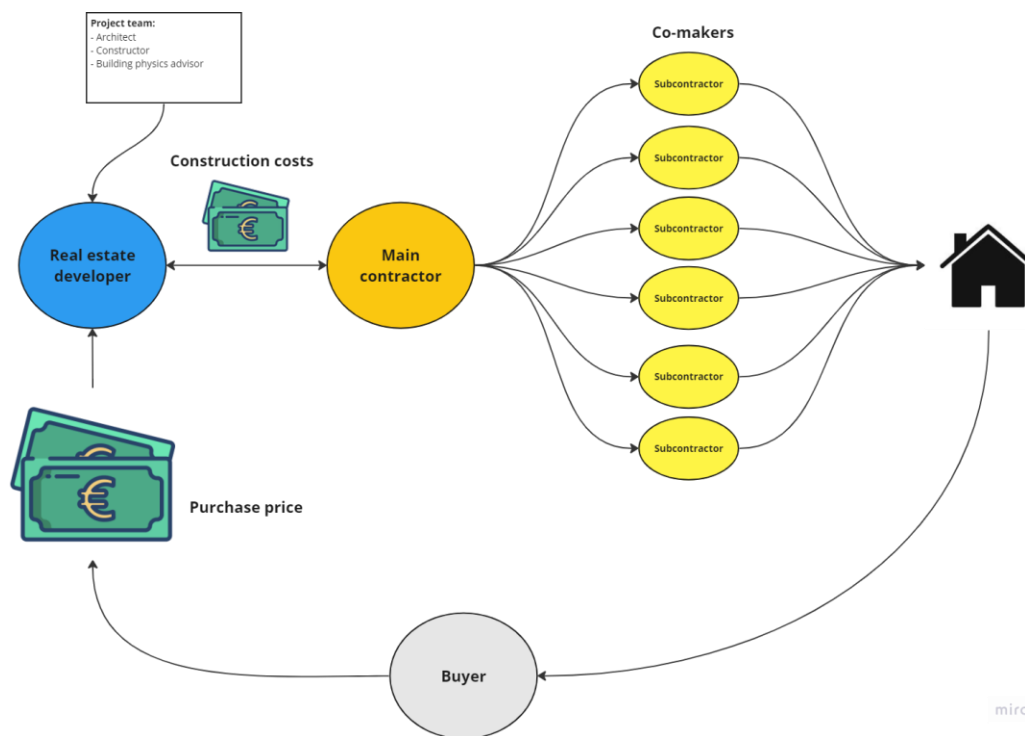


Figure 8 A schematic overview of the construction sector (Interview 9)

Now the reason this chain-like structure can become a barrier to the adoption of process innovations is that when a single party situated somewhere in the chain decides to automate and digitalize their processes this can have a direct impact on the other parties in the chain. This can best be illustrated using an example.

Consider the situation where the main contractor decides to optimize their processes through digitalization and automation. In this optimization, it is decided that they want a digital mockup of their project so that they can optimize the construction planning in terms of labor capacity, make a more accurate cost estimation, delivery planning, etc. The main contractor does not build the project on their own, as they receive parts of the project or elements from other companies, the sub-contractors. If these

sub-contractors do not want, or cannot deliver their products in a digital version to the main contractor, it becomes difficult for that main contractor to make a full digital mockup of the project, and thus receive resistance from other companies in their process innovation implementation.

Companies are aware of this resistance, upon getting asked what the biggest barrier to process innovations. To illustrate this resistance with a quote, “There are also various supplier and subcontractors already involved, because they also need to be part of that automation process ... you can want this as the only party in the chain, but you have to consider this integrally” (Interview 7).

The director in Interview 17 made similar remarks upon getting asked what the biggest barrier to process innovations as he answered. “The biggest problem does not lie internally within the company if you have a little bit of a change culture. But more because you’re just a small link, you don’t always influence the environment around you [referring to other companies]” (Interview 17).

Lack of technology awareness

The final barrier is the lack of technology awareness, described in the literature review as either having no awareness about the potential technologies that are present in the market (Iturralde et al., 2022), or the benefits and gains that can be obtained from these innovations are unclear (Demirkesen and Tezel, 2022). This barrier was also mentioned by a strategy advisor for sustainable housing, upon getting asked which factors are perceived as barriers to the implementation of process innovations, she replied “Ignorance about automation and the implementation thereof. In terms of what we don’t understand well, we also find difficult to adopt” (Interview 13).

Interestingly, concerning the benefits of different process innovations (see section 4.4.1 Benefits), the majority of construction industry interviewees demonstrated awareness regarding the benefits associated with these innovations. Furthermore, these benefits were pretty consistent across these interviews. The lack of technology awareness seems to be more of a problem among the people who execute the work, so the technicians and mechanics rather than the decision-makers in the company. For example, in Interview 14 the business developer explained that mechanics showed resistance while implementing digital tablets. He later found out that this resistance originated in that they saw no reason for the implementation. Only after he explained the benefits that the technology could bring to mechanics, did this resistance change (Interview 14). In a similar example, the Director of the total installation company explained that other companies working on the same project sometimes create resistance because they are not familiar with their prefabrication approach and therefore do not understand it (Interview 17). Lack of technology awareness thus forms an indirect barrier to the decision-maker, because it increases the resistance to change of their workers or other companies in the chain.

4.3.3 Perceived Behavioral Control – Barriers Construction Industry

Cost of implementation

This barrier, highlighted in the literature review, was also recognized in the conducted interviews. The CEO of a company that creates customized solar façades mentions costs to be the number one barrier to the implementation of automation and digitalization (Interview 11). An important dimension of this barrier is that small- to medium-sized enterprises struggle significantly more with this barrier. Relatively speaking the costs to implement innovations is larger for SMEs in comparison to large multinational companies (Interview 11).

But costs directly related to the adoption of innovation, such as upfront costs, are not the only dimension of the cost of implementation barrier. To implement innovation into your existing processes, a team

needs to be created that is made responsible for the execution of the innovation project, and that also costs money (Interview 7). Once this team is made available you are not done, however. In the construction sector, projects have different stages that they go through, roughly speaking from design to realization. And as the project manager of a contractor company (Interview 7) explains, when you brainstorm about digitalization and automation everyone is capable of drawing up a future vision that would be ideal. However, the actual implementation of that vision, so cutting up the vision into smaller steps that eventually lead to the realization of that vision is difficult. This is because “our process lead times are quite long, right. Before you go from a design to a proven case, that might take you anywhere from 3 to 4 years” (project manager, Interview 7). In the meantime, a lot can happen to the project.

For example, another frequently mentioned problem is innovation budget cutting (Interview 13, 14, 20). This refers to the situation where the board of directives no longer supports the ongoing innovation projects because of for example financial downtimes, or internal company pressures such as resistance from the workforce (see resistance to change). In this situation, the board decides to, sometimes temporarily, put innovation projects on hold. But this is destructive for the progress of those projects. This is because, in large corporations, innovation requires proper planning operations to make sure there is enough capacity in terms of money, but also capacity in terms of making sure the right people have time to work on the project. Budget cutting scraps the innovation project from the planning of important departments such as IT departments, necessary for digitalization (Interview 13 and 14). Once the innovation project is of the planning, it is hard to get it back on the planning, and substantial progress on the project is lost.

Lack of skills

This capacity problem is tightly linked to the skills, or rather the lack thereof, that you need to successfully implement those process innovations. This barrier was also discussed in the literature and refers to that the implementation of process innovations often requires highly educated workers. This barrier was also found in the conducted interviews. For example, the CEO of the customized solar façade company even stated that “what you see with automation, digitalization, and application of artificial intelligence, is that the work that is still truly necessary requires higher education ... process innovations determine the quality of the employees that you are searching” (Interview 11). However, the majority of employees either lack these skills, or there is only a limited amount that possesses the skills to successfully adopt and implement the process innovations. The notion of Newman et al. (2021) that companies hesitate to implement innovations because they require highly qualified personnel, thereby decreasing the production of the company was also discussed in Interview 17 and 20.

The dichotomy, between innovation and production was mentioned by the Director and owner of a family installation company (Interview 17). “Are you more of an early adopter or are you more situated towards the back? [refers to innovation adoption curve] ... Companies that are more situated towards the back, just want to focus on production and those are less focused on innovation” (Interview 17). In Interview 20 the sales director also recognizes how important the limited pool of talent is for successfully implementing process innovations. The problem, he elaborates, arises “once a big project comes in, then those smart heads [refers to highly qualified personnel] immediately get pulled away [from the innovation project]” because big projects lead to revenue (Interview 20). And this makes it “difficult to create capacity for people [refers to highly qualified personnel] as well as maintaining this capacity” that is necessary for the implementation of process innovations (Interview 20).

4.4 Drivers to Process Innovation in the Construction Industry

This section discusses the driving factors that influence the attitude, subjective norms, and PBC of construction industry decision-makers toward the adoption of process innovation. A conceptual overview of the drivers can be seen in Figure 9.

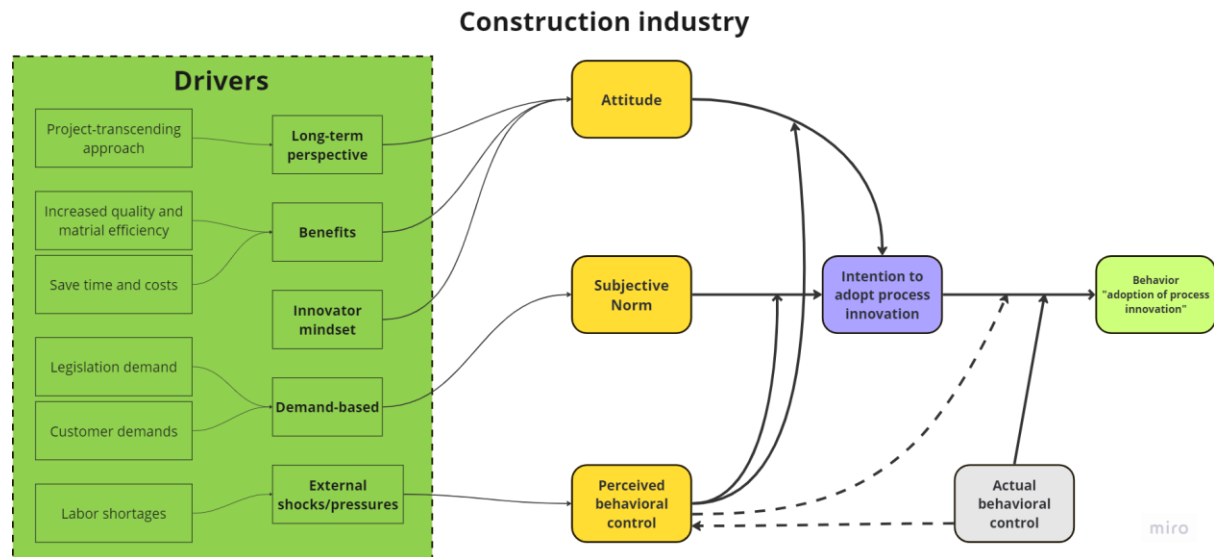


Figure 9 A schematic overview of drivers in the Construction industry

4.4.1 Attitudes – Drivers Construction Industry

Long-term perspective

In direct contrast to the short-term perspective attitude barrier is an attitude that links the adoption of process innovation toward a long-term outcome goal.

Project transcending thinking

Remember that the short-term perspective was embedded in the project-based approach that is representative of the construction sector. Thinking in single projects makes it hard to find the standardization that is necessary for process innovations such as digitalization and automation. It is therefore that in contrast to this short-term project-based approach, innovation adopters in the construction sector recognize this problem and look for a more long-term project-transcending or process-based approach (Interview 7, 21). This is portrayed by the director (Interview 21) of the company that assembles houses in a factory.

“Well, okay, and that means that you have to start thinking about whether you keep building more halls and stick to fairly traditional methods, or do you eventually seek a sort of shift in your business processes? From constantly reinventing the wheel, which construction is actually very good at, to something more like a mindset that can be applied more frequently, so not project-based thinking, but project-transcending thinking. This also means that the time and energy you invest can be used more often, thus it can pay off” (Interview 21).

In this long-term perspective, short-term profit is not sacrificed because the company also gains from it. Upon asking the director of an installation company why he is willing to take a 6% profit margin when he could get 8%, he answered “Continuity, our company has existed for 130 years” (Director, Interview 17). And it is through this focus on continuity, that the adoption of process innovations is

linked to outcomes in different fashion. With the goal of continuity, the adoption of process innovations functions as an enabler and is labeled positively in a variety of ways. For example, in preventing employees from leaving the company, “are you prepared to invest in people? In training and education ... So investing in innovations results in retaining people within the company” (Interview 17). But also in attracting new talent, because they “like that you’re doing it in that way [refers to an innovative approach] and so you also become an attractive employer” (Interview 17). This is in line with multiple interviews that stated that investing in process innovations is important to stay attractive for employees (Interviews 15, 17, 19).

The director (Interview 17) also recognized that this strategy does not work for maximizing profits in the short term, but that is not seen as a problem, because process innovation is not linked to the outcome of short-term profits but to that of company continuity, and “you’ll see that yourself, if you postpone for too long [investing in innovations], you’ll eventually cease to exist” (Director, Interview 17).

The contrast between the long- and short-term perspectives towards the adoption of process innovation is striking and is portrayed clearly by a quote from an Innovation process leader of a construction renovation company (Interview 15). His function is relatively new in the company and in explaining how his function was created the different attitudes are portrayed. “The company always neatly reserved an innovation budget of 300,000 euros because we need to innovate. And by the end of the year, it hadn’t been used. In fact, the budget was almost completely intact. The people from finance reacted like ‘this is great we have saved 300,000 euros’, and the CEO stated ‘no actually, we have stagnated’” (Interview 15).

Benefits

Next to having a long-term perspective, various factors influence attitudes toward adopting process innovations. These factors encompass the perceived and experienced benefits associated with the adoption of process innovations. This section will discuss the most frequently mentioned benefits that process innovations such as automation, digitalization, robotization, and prefabrication bring.

1. Increased quality and material efficiency

The first benefit that process innovation brings to the construction industry is the increase in quality and material efficiency of the delivered products and services. Take prefabrication for example. By automating construction processes and assembling elements and modules in a controlled workshop environment rather than on-site, one increases the control over the assembly phase. This controlled and conditioned setting brings numerous advantages, including independence from adverse weather conditions that may impede assembly progress (Interview 17, 18, 21), mitigation of disturbances within urban areas (Interview 7, 21), and improved oversight of the assembly process, leading to greater precision in work and more efficient material usage, both adding to an increased quality of the product or service delivered (Interview 13, 14, 19).

This last one was underscored by the director of an installation company (Interview 19). She recounted how, before digitalization and prefabrication, engineers were constrained to cramped spaces within buildings, relying on drawings and on-the-job problem-solving. This often resulted in incorrect installations and non-standardized practices, with instances of serious errors such as connecting cold water pipes to warm water pipes, leading to substantial energy losses (Interview 19). To prohibit these mistakes, the company established a prefabrication factory and utilized digital tools for the design of the systems. They developed modular solutions that could only be connected as intended, ensuring correct installation. Consequently, the company can now deliver higher-quality products to clients, supported by a digital process that accurately determines material requirements, thereby also minimizing waste (Interview 19).

2. Save time and costs

Another frequently mentioned benefit of implementing process innovations regards the saving of time and costs (Interview 8, 12, 13, 14, 18, 19, 21). Before explaining this further, it is important to note that these cost and time savings arise only once the process innovation has been successfully implemented in the company.

To illustrate how digitalization can save time in a construction process, we turn to the following explanation of an architect who works for an innovative digital company that makes parametric designs for projects that incorporate “boundaries, constraints, variables, and parameters that are within a project ... we actually see what are all the variables there are within the KPIs of a project to be able to, to like create the best possible projects” (Interview 12). In the early stages of the construction process, the design needs to adhere to a substantial amount of standards called building permits. This is done to make sure that the projects comply with building codes, safety standards, legal compliance, quality assurance, etc. Now the software of the company allows real estate developers to find an optimum version of the project that considers a wide range of variables. They can adhere to these quality standards and legislation at a much faster pace. What architects would traditionally take 8 months, the company can now do in one to two months, meaning you already save 6 months’ worth of time and resources (Interview 12). The following quote can be used to illustrate the effectiveness of this method and how much time and cost it saves in comparison to traditional architecture. “You cannot evaluate (i.e. charge) us on how much time we spend. So usually architects charge by the hour. We cannot charge by the hour because we do things too fast. Thanks to technology, right.” (Interview 12).

In another example, the director of a housing assembler, that uses a digitalized and automated factory to assemble houses similar to a car factory, explains how a process-based mindset that was necessary for digitalization now means that they do not “have to reinvent the wheel” for every project, thereby also saving time (Interview 21). The same benefits were mentioned by the director of a large installation company (Interview 19), who stated that once the process is digitalized and automated the process becomes simpler because the projects now also have similarities because of standardization and modularity. This also creates benefits in the long run, because when you have to provide maintenance for the installation systems that you installed 10 to 30 years ago, it makes it a lot easier when the different systems share similarities (Interview 19).

Process innovations can also play an important role in optimizing the maintenance and management of buildings. The circularity manager of an engineering bureau (Interview 8) explains how a process innovation significantly saves time by leveraging smart sensors and custom scripts for remote monitoring. Previously, five employees monitored around 200 buildings, using predetermined plans. This sometimes caused a mechanic would visit a building and a few hours later something would break down (Interview 8). However, by innovating this process using smart sensors, the company now remotely monitors all buildings in real-time, proactively addressing issues, thereby being able to only need 2 people for those 200 buildings instead of 5. Similar benefits were reported by another maintenance and management company (Interview 13, 14).

Innovator Mindset

Finally, a similar innovator mindset can be found in the construction industry as was discussed for the restaurant industry. This describes the attitude of a decision-maker who is constantly looking to optimize, improve, and innovate existing processes. To these individuals process innovations serve as a means to an end, that is the process innovation is linked to the desired outcome of constantly improving the existing situation. This is illustrated by the following quote of the Director in interview 19, “I mean, we are always looking for ways to improve. How can it be more efficient? How can we make it cheaper, how can it be faster? That’s what drives us and what we find really enjoyable”

(Interview 19). A similar attitude was found in the Director and owner of the firm that assembles houses in a factory. when asked how he came up with the idea he responded “Well, I am someone who thinks every day that what is good today can be better tomorrow” (Interview 21). And it is not that everything went smoothly once he decided to continue this plan. “I also think that half the world has declared me crazy. They said it will never work in construction, because construction is used to building a prototype every time” (Interview 21). These types of decision-makers are suitable leaders for the implementation of process innovations because their positive attitude toward process innovations and their capabilities for improvements create the stability that these difficult implementations go through.

4.4.2 Subjective Norms – Drivers Construction Industry

In this section, we discuss how referent others can drive the adoption of process innovations. This is done through what could be summarized as demand-based process innovations

Demand-based

Demand-based refers to how the adoption of process innovations is influenced by external parties that can drive a company to adopt process innovations. In the conducted interviews it became clear that the source of the demand can come from roughly two sources. Firstly through regulatory bodies that implement legislation that force certain companies to adopt a process innovation to be able to adhere to the legislation standards. Secondly, customer or client demands can force companies to adopt process innovation to meet the demand. Both will be discussed.

Legislation demand

In the first example of demand-based innovation, the need for companies to adopt a process innovation resides in that it is mandated by legislation. A good example was discussed in Interview 8, in which the business circularity manager discussed that “legislation is now coming from Europe requiring companies to report on how sustainable they are” (Interview 8). He referred to the CSRD law or Corporate Sustainability Reporting Directive. He mentions how companies will now have to report on their sustainability practices in the same fashion that everyone is obliged to do for financial reporting. For the CSRD companies have to set up visions and goals regarding sustainability and also how they are monitoring those goals. “But this means that you are going to need data ... So, it’s just becoming more and more important for us to digitalize and automate. ... Without that data, your accountant can’t sign off, and you haven’t fulfilled your reporting obligation.” (Interview 8).

A similar example was mentioned by the Director and owner of a roofers company (Interview 18) that decided to make a customized digital application in reaction to the Quality Assurance Act (Wet kwaliteitsborging) (Interview 18), in which the government demands more supervision and control in the construction sector so that builders adhere to the applicable regulation (Rijksoverheid, 2024). The application allowed the director to follow a stepwise digital approach that requests photos per the required control step (Interview 18). In Interview 13, the sustainable housing strategy advisor discussed another regulatory demand, called the Recognized Energy Saving Measures Lists (EML, Erkende Maatregelenlijsten energiebesparing). The advisor discussed that you need to investigate how you can make your buildings more sustainable, which could be changing lighting, insulating the roof, but also the digitalization of your building management system (Interview 13). “And that’s even considered a recognized measure, so essentially the government also says, you must ensure that you have a building management system that is remotely readable [referring to that it has to be smart]” (Interview 13).

This demand-driven process innovation can serve as a significant motivator, even for companies less inclined to innovate their processes. For example, a company mainly invested in digitalization and prefabrication in their construction processes, yet less focused on sustainability, may not be leading in

that area, “but we will just follow the legislation” (Interview 17). Thus the companies adopt whatever process innovation is necessary to adhere to regulations.

Customer demands

Next to demand from the government, clients and customers can also place a demand on companies that can force them to adopt process innovations (Interview 13, 14, 21). One example regards the maintenance of the building, as explained in Interview 13, in which the client demanded more insights into the work that was performed, referring to for example how many malfunctions were handled and when, how much time it took to respond (Interview 13). In reaction to this demand, the company would have to better structure its internal work processes.

“So basically, well, the driving force behind this development [digitalization] was the customer. They wanted more digitization because they sought greater insight and preferred having everything in one place rather than scattered across various paper files. So that was the driving force, and then it was like, okay but then we also need to improve our basics surrounding our own workflow, and that’s what we started working on” (Interview 13).

In a similar example, the Director of a housing assembly company (Interview 21), came up with a vision to digitalize and semi-automate their assembly line of houses. In doing so, they also demanded from their suppliers that they digitally deliver their products as well (Interview 21). “yes, and then you say to your co-makers, with whom you collaborate extensively, ‘we believe that you should also draw the installations in Revit [modeling application]’, well that is happening now, and ‘we believe that the galleries we acquire from you should also be in Revit or at least be delivered via an IFC file [BIM type file]”(Interview 21).

Similar client-demand innovation is happening in tender markets. Tenders involve companies competing for contracts or projects. In Interview 14, it is described how in the maintenance and management of buildings, clients issue new tenders every x years. Over time, clients observe companies implementing process innovations that they appreciate and incorporate these into subsequent tenders (Interview 14). So with every tender iteration, all competing companies are required to incorporate those innovations if they want to have a chance of winning.

It is important to note that this demand for certain process innovations, such as digitalization can also be challenging for some companies, especially smaller SMEs. For example, in Interview 11 the CEO mentions that the client requested his company to make a BIM object of their product. His response is striking “Just a BIM object? Tailored to incorporate the new solar cells, colors, and everything else, we’ll have to come up with that on the fly, and every time that systems get updated, we should update our BIM objects as well” (Interview 11). This is simply too costly for them.

4.4.3 Perceived Behavioral Control – Drivers Construction Industry

External shocks: labor shortages

The majority opinion among the participants in this study says that labor shortages can act as a driver for the adoption and implementation of process innovations (Interview 7, 8, 9, 14, 15, 17, 18, 21). The director of a family installation company called the relationship between labor market shortages and investing in process innovations “one to one” (Interview 17). Furthermore, the project manager of a large contractor firm calls process innovations as a remedy for the lack of capacity in terms of workers “the most important driver for our sector” (Interview 7). Similarly, the designer of the company that develops a bricklayer robot, explains that the company cooperates with a launching customer who sees the investment in the robot as a way to make sure that they can maintain production in times of labor shortages (Interview 16).

In another example, the circularity manager in Interview 8 notes that as labor shortages increase, unprepared companies may be forced to outsource more of their work. By proactively investing in process innovations and ensuring that they have efficient business operations, his company can handle a larger volume of projects (Interview 8). The business developer in Interview 15 also stated that the availability of workers will only decline, while the work at hand will only increase, thereby believing that optimizing business processes so that workers can do their work more efficiently is the “key to success” (Interview 15). Also interesting to note is that one business developer thought it was an interesting perspective to use labor shortages as an argument to create more support for the adoption and implementation of process innovations in his company (Interview 14).

However, not all opinions are the same. The director of a large installation company said that she spoke to some more traditional installation firms to which she presented their prefabricated digitalized approach to making standardized installations. She said that the owner responded with “I don’t like that at all, because what will there be left for my own people to do?” (Interview 19).

Then there are also interesting additions to the relationship between labor market shortages and process innovations. For example, the CEO in Interview (11) discussed that once process innovations are implemented, the type of worker that you need also shifts. He explains that the work that remains is the work that requires a higher education, which will create new problems in the long term because the work that processes innovation usually takes over that of lower-educated jobs. Furthermore, the more highly educated people that are necessary for implementing these process innovations are hard to find (Interview 11, 19). It is here that there is some friction in the pressure that labor shortages put on companies to implement process innovations. On the one hand when you digitalize and automate processes you save labor, but on the other hand you will need to find more skilled employees which are also hard to find and that will probably remain the same for the future. The Director of a large installation company therefore also mentioned that the adoption and implementation of process innovations are more driven by societal goals such as the energy transition because it increases the speed with which the company can work towards these goals (Interview 19).

5. Discussion

This section will reflect on the analysis of the results by first highlighting the over-coupling similarities and differences between the drivers and barriers in both industries. These similarities and differences make up the factors that ultimately influence the decision-maker's decision to adopt the process innovation. The generalizability to other sectors will then be discussed. This section ends by discussing the limitations of this work and highlighting directions for future research.

5.1 Similarities between Industries

Innovator Mindset

The first similarity is that in both industries there were participants who demonstrated an innovator mindset attitude toward the adoption of process innovations. Decision-makers who demonstrate this attitude are characterized by a continued search for improvements to the existing situation (Interview 2, 19, 21). The benefits that come with the adoption of process innovations are often in line with this search for improvements, as many benefits include improvements in productivity, quality, and time savings (Interviews 2, 4, 5, 7, 8, 12, 13, 17, 18, 19, 21). Because their goal is to continually improve processes, their behavioral beliefs link the adoption of process innovations to the outcome of benefits and improvements thereby creating a positive attitude toward the adoption of process innovations. Since this research did not control for individual characteristics in terms of age and education, it is hard to determine what causes an individual to adopt or create this type of attitude towards constant improvements.

Necessity

The second similarity between the construction and restaurant industries is the role that necessity plays in the adoption of process innovations. As a general rule the higher the necessity the more likely the decision-maker adopts the process innovation. Necessity can have its origin in a variety of factors that can either increase or decrease necessity. For example, in both the construction and restaurant industries necessity is disrupted by resistance to change from employees or suppliers and clients (chain dependency) influencing the subjective norm of the decision-maker. If the employees of the company actively resist the implementation of a process innovation, the necessity to adopt and implement decreases. In addition, for the restaurant industry, customers can also influence the subjective norm in the case they do not like the process innovation this has similar negative effects to the necessity of adoption. Necessity, or the lack thereof, as a barrier to the adoption of process innovations, is also in line with Zegel and Cloosterman (2021) who found that a lack of urgency and priority act as a barrier to the adoption of process innovations.

Necessity can also be increased by a variety of factors. For example, both the restaurant and the construction industry can experience external shocks that create a necessity for the adoption of process innovations. In the restaurant industry, this became evident when QR codes were adopted to continue operations of the restaurant under the pressure of COVID-19. But also labor shortages, which can create a necessity among companies to look for alternative approaches to existing business processes. Take for example the bricklaying robot, which was developed by the company in Interview 16 because the launching customer recognized the need for such an innovation because of labor market shortages. In the construction industry necessity can also be increased by external demand. This was visible in the form of government legislation, requiring companies to adopt certain process innovations to adhere to regulation. But also in the demand companies pose on each other, such as a main contractor that demands products from suppliers in both the physical and digital form, creating the necessity to digitalize among sub-contractors (Interview 21).

The attitude of the decision-maker can also play an important role in the creation of necessity. For example in the restaurant industry when decision-makers see no added benefits to the implementation of process innovation (Interview 3), or see it as disruptive to their restaurant's value creation (Interview 4, 6). This disrupts the necessity necessary for the adoption of process innovations. In the construction industry, necessity can also be disrupted by negative attitudes that do not link the process innovations to desirable outcomes. This was discussed in Interview 19, where the Director shared a story of an installation company that would not adopt prefabrication and digitalization because they had no idea what there would be left to do for its employees. These examples demonstrate the importance of the linkage that the decision-maker has between the behavioral belief and the outcome, meaning what will be the result of the adoption of a process innovation, and is this result favorable? If a decision-maker has a negative association with the outcome, as in the example, the negative attitude will disrupt the necessity for the adoption.

Resources and Opportunities

The third similarity relates to the factors that influence the perceived behavioral control (PBC) of decision-makers. This refers to whether the decision-maker has access to the requisite financial resources and opportunities (Ajzen, 1991). If not, this directly influences the ability of the decision-maker to adopt the process innovation. This is true for both the construction industry and the restaurant industry. For example, if the director of a main contractor company has a long-term perspective that views the adoption of process innovation as something valuable (attitude), and the employees of the company are also in favor of the process innovation (subjective norm). His company might still not be able to adopt process innovations when the company does not have the financial resources to cover the implementation costs. This also holds for other factors influencing the PBC. Concerning infrastructural barriers in the restaurant industry, some restaurants may have too narrow aisles between dinner tables, making it impossible for a serving robot to pass (Interview 5). Other decision-makers in the restaurant industry might want to automate processes in the kitchen but are unable to find robots that can meet the speed necessary during peak hours (Interview 1).

This aspect is also linked to characteristics of the company such as its size. SMEs often have less financial resources, making the upfront cost barrier more problematic (Interview 11, 15, 20). Note that the decision-maker in Interview 20 is part of a large company but explicitly stated this problem for SMEs. In contrast, large companies have difficulties with bureaucracy, which slows down progress and disrupts the agility necessary for implementing process innovations (Interview 13, 14). Restaurant chains that require process innovation to be suitable for all individual restaurants could also face this agility problem. In some cases, medium-sized companies seem to get the best of both worlds concerning these two challenges. They have enough financial profit to cover the cost of adopting process innovations while keeping the agility necessary to successfully implement them (Interview 17, 20).

Applicability

In the analysis, it became clear that some process innovations are more applicable for certain companies than for others, this was evident in both sectors. In the construction industry, the type of process innovation adopted often depends on the type of work that the company performs. Companies that focus on the management and maintenance of buildings will not adopt BIM as they work with existing structures. Therefore these companies drift more towards sensory technology to create a digital picture of the performance of the building in terms of energy and comfort and display this to their customers through dashboarding. In contrast, new construction has a lot of potential for the implementation of BIM and prefabrication, as these companies have the chance to design the structure from a digital file up until the actual product. In the restaurant sector, this applicability between the company and the

process innovation is also present. Restaurants that lean a bit more towards fast and lower-cost food, are more open to automation and digitalization of the restaurant operations in both the kitchen and food serving (Interview 1), which is in line with the findings of Grobbelaar et al. (2021). While restaurants that strategize more on personal contact with their customers are less likely to adopt robots (Interview 4 & 6).

5.2 Differences between industries

Time horizon

An interesting difference that is evident from the results is that in the construction industry, there is a clear distinction in the decision-maker's time horizon perspective, while this did not arise from the restaurant interviews. Before discussing why this difference occurred, the time horizon perspective is first explained. The time horizon refers to the length of time, or a fixed point of time in the future at which the value of the process innovation is evaluated. In the construction industry, we saw a clear distinction between the long-term and short-term perspectives that shape two completely different attitudes toward the adoption of process innovation. This is because the opportunities and challenges that come with process innovations are evaluated differently depending on the time horizon with which someone looks at the adoption.

To illustrate, in the case of a long-term perspective, the decision-maker looks at the implementation of process innovation over for example 5 years. He or she is aware that the implementation might take three full years before it is successfully integrated into the processes of the company. In these three years, the implementation costs are high, meaning that it will significantly decrease the profit margins of the company. But the process innovation also comes with benefits once implemented. The innovation is top-notch and will attract high-skilled workers who are eager to work with the newest technologies, which is good for the continuity of the company and helps with finding the talent that the labor market is running low on. Once implemented the innovation also helps with increasing the productivity of the company, meaning that the same demand can be met with fewer people thereby saving on labor costs. Automation also decreases the error margins in production, meaning that the company produces fewer failed products. This means material efficiency increases which is good for sustainability while also decreasing the amount of material needed, again lowering the costs. In addition, the long-term perspective can also help with the resistance from chain dependency, because when companies collaborate on longer-term projects, it creates predictability for the involved partners which allows them to look at ways to standardize their processes and implement process innovations. The project manager in Interview 7 explains that you need “a point on the horizon”, so a project that is ambitious in both scale and duration, and that involves a substantial amount of revenue so that it can attract subcontractor collaboration.

In contrast, if the decision-maker has a short-term perspective of say one year, then the three-year implementation phase in which cash is spent with no added benefits, will immediately decrease the interest of the decision-maker to adopt the process innovation. The short-term perspective as a barrier to the adoption of process innovation is in line with the findings of Zegel and Cloosterman (2021), which stated that the short-term focus causes companies to perform limited investments in process innovations.

This clear distinction did not show in the restaurant sector, first of all, because the business operations of the restaurant sector are naturally more short-term. In broad terms the daily operations are the same, creating value for the customer through food, drinks, and a hospitality experience which have to be created every day. Throughout the year some days are busier, such as weekends and holiday periods, but the value you create is the same. The similarity of the operations creates a natural short-term

perspective, in which the decision-maker has to keep the restaurant running daily. This means that there are fewer long-term goals, rendering the need for a long-term perspective unnecessary. This is where the construction industry's situation is different because it is faced with a variety of societal goals that play a role in both the present and the future, such as solving the housing crisis and playing an important role in sustainability transitions. For example, the housing crisis requires companies to produce houses at a faster pace meaning that existing business processes have to become more productive to meet this new demand. A long-term perspective can help optimize and do this as explained earlier. On the contrary, the restaurant industry is not pressured by a long-term societal goal requiring restaurant decision-makers to adopt a long-term perspective.

Human Capital

Another difference between the two sectors was the approach to human capital. The literature review showed that for both sectors human capital-related issues formed a barrier to the adoption of process innovations. However, in the analysis, the lack of skills barrier seemed to have more impact on adoption in the construction industry than in the restaurant industry. Decision-makers in the restaurant industry discussed that employees did not have the skills or knowledge to develop process innovations, but they did not expect them to have these skills (Interviews 2 and 6). Rather the decision-makers expected the process innovation to be easy to implement. In contrast, process innovations in the construction industry do require high-level skills from the employees of the company, because the adoption of C4.0 technologies is complex and often requires in-depth knowledge to effectively implement them (Interviews 11, 14, 17, 20). The lack of these specialized skills is in line with the findings of Siriwardhana and Moehler (2023), who stated that the overlooked importance of specialized skills hampers the potential of C4.0. Furthermore, the lack of employees with the right skills was also mentioned as a barrier to adoption, because it takes away from the production capacity of the company (Interviews 17 and 20). This is in line with Zegel and Cloosterman (2021) who stated that companies have trouble with creating the internal capacity necessary to implement the process innovation. The amount of people who hold these specialized skills is scarce. This makes the adoption of process innovations in the construction industry difficult for some companies because it means that the more process innovations you adopt the more specialized skills your workers need (Interview 11, 19), and again these people are hard to find.

5.3 Generalizability to Other Industries

This section will briefly discuss the generalizability of the identified factors that influence the adoption of process innovations in sectors outside of the restaurant and construction industry. The factors that can be used as a framework to investigate barriers and drivers to the adoption of process innovations across different industries are the *Innovator mindset*, *Necessity*, *Resources & Opportunities*, and *Applicability*. First of all, these factors were identified in both the restaurant and the construction industries, which are two completely different industries in terms of characteristics and value creation, showcasing their wide relevance. Secondly, these factors were supported by the literature review performed in this study.

In contrast, *Time horizon* and *Human Capital* were identified to differ between the construction and restaurant industries. Although human capital-related barriers were mentioned for both industries in the literature review, the decision-makers in the restaurant industry did not expect or require their employees to have these skills. Finally, the *Time Horizon* of the decision-maker is a generalizable factor in industries that involve the use and development of technologies, such as manufacturing, or sectors that are involved in longer-term societal goals, such as sustainability-related transitions. The short-term

focus was also mentioned as a barrier to the adoption of process innovations in the manufacturing industry in the work of Zegel and Cloosterman (2021), which makes sense as C4.0 types of technologies have a large familiarity with manufacturing type technologies. When construction moves more toward digitalization and prefabrication the industry also becomes more similar to the manufacturing industry. On the contrary, the *Time Horizon* does not influence the decision-makers choice to adopt process innovations in industries that have more day-to-day operations focus, such as the hospitality industry, and the culture, sport, and recreation industry.

5.4 Limitations

This research has some limitations that need to be addressed. First of all, as already discussed in Section 4.1.1, the barrier lack of trust was not present among the participants in this study. It was explained that this barrier is of more concern to the customer, and therefore it might not been mentioned by the decision-makers included in this study. A second limitation of this research is that the focus might have been too broad regarding process innovations. On the one hand, this creates a wide understanding of the factors that influence the adoption of process innovations, while on the other side, it is challenging to obtain actionable and detailed best practices for specific process innovation implementations. A third limitation is that this study did not control for individual characteristics such as background level education and age. There could be a correlation between someone's education or type of degree and their attitude toward technology and process innovations. A fifth limitation is a general one of qualitative interviews, which provide self-reported data and can thus be subject to bias, as well as motivational agendas (Rowley, 2012). Regarding self-reported data, when the participant said that process innovations increased the productivity of the company, this study had to believe this to be true because it did not verify the productivity increase with actual quantitative data. Furthermore, regarding biases, this study did not control for potential secondary roles or side businesses among the participants in the study. This could create a bias in the case that there was someone in the sample who was a provider or developer of a certain process innovation or technology for which they conveyed a particular narrative around that innovation (Roberts et al., 2021).

5.5 Future Research

Future research could look into the perspectives of workers who in the end execute the work and interact with the process innovations. These perspectives could generate critical insights that could create a better understanding of best practices from the worker's perspective. Secondly, future research could apply a similar research setup but include only one or two types of process innovations to be studied. This could create more actionable and specific knowledge. Thirdly, further research should also incorporate the educational background of the decision-maker as well as other variables such as age that could influence someone's perspective on process innovations. Finally, this research shows that companies in the construction industry deal with a complicated architecture of the branch, referring to the chain-like organization. Future research could investigate using a more ecosystem framework how these chains can best be organized so that individual companies can situate themselves optimally concerning other companies in the chain. This way the entire ecosystem could be optimized with regard to the production of houses to combat the housing crisis, but also to spur the adoption of process innovations to deal with the pressures of the current and future labor market shortages.

6. Conclusion

6.1 Main Conclusions

This research was conducted to answer the main research question, *what are the factors that influence the intention and decision to implement process innovation among decision-makers in the restaurant industry and construction industry?* To answer this question, this research conducted qualitative research by interviewing decision-makers from two different industries, using questions based on the Theory of Planned Behavior. The sectors included the restaurant business and the construction industries. From the results, it became clear that a variety of drivers and barriers can influence the adoption of process innovations by shaping attitudes, subjective norms, and perceived behavioral control (PBC) toward either an increase or a decrease in the likelihood of adopting the process innovations. From these barriers and drivers, the main factors were identified influencing the adoption, four of which applied to both industries and are therefore generalizable to other industries, and two specific to the construction industry. The main factors that apply to both industries are the following *Innovator Mindset*, *Necessity*, *Resources and Opportunities*, and *Applicability*. The two other factors that were only applicable to the construction industry are the *Time Horizon* and the *Human Capital* factor.

In both the restaurant and construction industries the adoption of process innovations is influenced by the following factors. First of all, some decision-makers have the *Innovator Mindset* attitude, which describes a certain positive attitude toward the adoption of process innovations because they are constantly looking to improve existing processes. The benefits that originate from the adoption of process innovations, such as increased productivity, quality, and time savings are in line with this constant search for improvements, which thereby increases the likelihood of the adoption of process innovations. Secondly, the *Necessity* to adopt can be increased by external pressures that drive decision-makers toward adopting process innovations, for example, because of COVID-19 or labor market shortages. In addition, *Necessity* can also be increased by factors that influence the subjective norms of decision-makers in the form of legislation or customer demand that requires the adoption of certain process innovations. In contrast, *Necessity* can be decreased by resistance to change from the employees of the company, or if the process innovation adds no value to the existing business operations. Thirdly, *Resources and Opportunities* play a direct role in influencing the PBC of decision-makers, which means that barriers and drivers that influence the PBC can directly influence the decision-maker's ability to perform the behavior. Barriers that influence the PBC are for example cost of implementation, infrastructural barriers, or technologies that are unable to meet the company's requirements. Finally, *Applicability* refers to the process innovation being suitable to the type of company and its business operations. In the restaurant sector, for example, fast food restaurants have customers who are less bothered if the restaurants use robotization and digitalization in the creation of their meals. In contrast, restaurants that aim for a more personalized approach are less likely to use robots in their restaurants. Similarly, in the construction industry companies that are in the business of management and maintenance of buildings are more likely to use sensor-based technologies to create a digital picture of the performance of the building in terms of energy and comfort and display this to their customers through dashboarding. In contrast, new construction has a lot of potential for the implementation of BIM and prefabrication, as these companies have the chance to design the structure from a digital file up until the actual product.

Additionally, for the construction industry, the *Time Horizon* and *Human Capital* also influence the adoption of process innovations. The *Time Horizon* refers to the type of time perspective that the decision-maker has concerning the adoption of process innovation. A long-term perspective is more likely to lead to the adoption of process innovation because it incorporates the implementation challenges and places them in the perspective of a higher goal, such as company continuance. Finally,

Human Capital relates to the specialized skills that are required for the adoption and implementation of process innovation in the construction industry. These highly skilled workers are often scarce in a company, therefore it is often difficult to build the capacity necessary for the implementation.

6.2 Policy Recommendation

For the restaurant sector, based on the findings of this research, there seems to be no evidence that justifies a government intervention related to process innovations. First of all, no legislation creates a necessity for restaurant decision-makers to adopt process innovation, this is visible in Figure 6 in which no driving factor can be seen that influences the subjective norm of a restaurant decision-maker. Furthermore, the adoption of process innovation in the restaurant industry is mainly dependent on the type of restaurant and the suitability of the process innovation (*Applicability*), the attitude of the decision-maker (*Innovator Mindset*), and whether they perceive the process innovations to bring benefits to their restaurant. These are individual characteristics and abilities of the restaurant, which justifies no government intervention.

In contrast, the construction industry faces a multitude of legislation pressures from the government that originate in societal goals such as climate ambitions. These ambitions create the necessity that forces companies to change their existing processes. However, the government should be careful that it creates a growing necessity to adopt the process innovations necessary to adhere to regulations without companies having the financial and human capital resources to do so. The technician shortage will likely only increase in the Netherlands because younger people opt less for a technology-related education (Heyma et al., 2022). This means that a large-scale transition might increase the necessity, while the bedrock of financial resources and human capital resources important to the success of these transitions is lacking. Especially since this research highlighted how the adoption of process innovations requires more workers who have specialized skills.

With a decline in productivity across sectors (van den Berg et al., 2022), increasing labor shortages, and the underutilized implementation of labor-saving process innovations (van Kesteren et al., 2023a and 2023b), this study, therefore, argues that there should be a government focus on making sure there is money invested in skills development necessary for C4.0 type process innovations in the Netherlands. This is especially applicable to small firms that are willing to adopt C4.0 type process innovations, but lack the financial resources and have trouble finding the necessary human capital. It is argued that government intervention is justified for increasing the human capital pool necessary for C4.0-type process innovations because it is part of the foundation of the government's transition goals.

Great examples of approaches to increase skills and awareness have been demonstrated by the work of Pomp et al. (2022 and 2023). As part of a larger digital platform strategy to increase and connect businesses, Pomp et al. developed a platform called IoT4H that helps craft businesses explore and implement IoT solutions in Germany. The government could stimulate the creation of similar platforms for the Dutch construction industry, related to other C4.0 type technologies such as BIM, sensor-based technologies, 3D printing, prefabrication, and artificial intelligence. This can improve the lack of skills and technology awareness because companies can educate their workforce using these platforms. This in turn decreases the resistance to change that affects the subjective norm of the decision-maker, thereby creating fewer barriers to the adoption of process innovations. The pool of talents that have skills in these areas needs to increase to enhance the adoption of process innovations in the construction industry, which is still important today as labor market shortages are predicted to remain a problem in the coming years.

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8. Appendix

8.1 Appendix A: Interview Guide

Semi-Structured Interview Guide

Date of the interview: XX-XX-XX

Author: Bas Buurman (student nr. 1244511)

Purpose: Master thesis Innovation Sciences

Actions before the interview:

Inform the interviewee and let them sign informed consent

Background information – to be gathered before the interview

Name of the person(s) interviewed:

Company:

Active in which industry/sector:

Job title:

Introduction from the interviewer:

- Introduce myself + provide information about my educational background.
- Explain the purpose of this interview: The Netherlands is dealing with labor shortages throughout different sectors. The hospitality and construction industries are examples of these shortage sectors. These labor shortages are predicted to remain a problem throughout the coming years, because of population aging and an already high labor participation rate. This could potentially threaten societal goals such as climate ambitions but also decrease overall economic prosperity. An important solution could be to increase labor productivity in the shortage sectors, thereby decreasing the demand for workers. One way of doing this is through the implementation of labor-saving process innovations such as automation, robotization, and digitalization. Most employers have neglected to invest in labor-saving process innovations explaining the stagnation of labor productivity. This research is trying to understand which barriers influence the adoption of process innovation and whether the current situation changes the perception and decision-making process of employers in terms of implementing process innovation.

Opening questions:

1. Can you please introduce yourself and provide some background about the activities of the company you work for?
2. What is your specific function in the company?
3. How many employees does your company have?

Awareness of process innovation options such as automation, digitization & robotization:

4. Are you familiar with process innovations such as automation, robotization, or digitalization?
If no: **show an example of process innovation (example depends on the sector)**
5. Has your company ever invested in or implemented process innovations such as automation/ robotization/ digitization?
If yes
 - a. What kind of investments or implementations in process innovation did you implement in the past
 If not
 - b. Which type of process innovation do you think would benefit your enterprise?

Implementation of automation/ robotization/ digitization and potential enablers:

a. Attitudes

- i. What are your perceptions of the benefits of implementing process innovations such as automation, and digitalization within your enterprise?
- ii. How do you personally feel about the idea of integrating automation, digitalization, and robotization into your enterprise's operations?
- iii. How do you think these innovations would impact (or impacted) the efficiency and effectiveness of your business processes?
- iv. How, according to yourself, do (or did) employees experience implementations of process innovation?
 1. What are concerns you have about your employees when thinking of implementing process innovations such as automation, digitalization, or robotization?

b. Descriptive norms

- i. How common do you believe it is for other enterprises in your industry to implement process innovations like automation, digitalization, and robotization?
- ii. To what extent do you feel pressure or encouragement from others in your industry to adopt these innovations?

- iii. Have you observed or heard about other enterprises successfully implementing similar innovations?

If yes:

- 1. How did that influence your perception of their feasibility and desirability?

c. Perceived behavioral control/

- i. What factors do/did you perceive as facilitating or hindering the implementation of process innovations such as automation, digitalization, and robotization within your enterprise?

- 1. How do/did you overcome these problems?

- ii. How do/did financial resources impact the decision and implementation of the process innovation?

If yes:

- 1. Could you share insights into the sources or support mechanisms that contributed to the financial backing of implementation or initiative?

- iii. How do the skills and abilities of the employees in the company affect the implementation of process innovations such as automation, digitalization, and robotization?

- 1. How were employees prepared or trained to work with the new process innovation?

Relationship between labor shortages and process innovations.

- 1. Have you ever experienced labor shortages, when?
 - a. If yes
 - b. How have labor shortages ever affected your operations?
- 2. Can you share your thoughts on the relationship between labor shortages and automation/robotization/ digitization?
- 3. How would (or do) labor shortages affect your intentions to invest and implement some form of process innovation?

Closing questions

- 1. Was everything clear?
- 2. Do you have anything to add?

8.2 Appendix B Ethical Review Approval

RE: Ethical review submission master thesis - Bas Buurman



Severens, Marjolein on behalf of Ethics
To: Buurman, Bas



Mon 12/02/2024 19:43

You forwarded this message on 13/02/2024 13:36.

Dear Bas,

Your application titled "Investigating the decision to implement process innovations to reduce labor shortages using the theory of planned behavior." (ERB20224IEIS8) has been approved by the ERB.

We assume that you have answered all questions correctly. We will perform regular spot-checks so you need to keep your documentation (ERB form, informed consent forms, surveys/interview questions, description of experiment/prototype etc.) available for at least 6 months.

Good luck!

Dear regards,

Marjolein Severens
ERB student assistant